

Decomposing Portmanteaux: Pronominal Morphology and Prosodic Structure in Kono

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Abstract

In many languages with STAMP (Subject-Tense-Aspect-Mood-Polarity) morphology, subject pronouns and tense-aspect auxiliaries surface as apparent portmanteau morphemes whose forms can vary quite unpredictably across paradigm cells. These patterns have often been treated as suppletive, effectively derived from dedicated morphological operations. This paper examines STAMP forms in Kono (Central Mande, Sierra Leone) and argues that the observed paradigmatic variation does not reflect suppletion. Rather, pronouns and auxiliaries remain distinct morphosyntactic objects throughout the derivation, with surface variation arising from independently motivated phonological processes. The same phonological processes that derive pronominal STAMP forms also apply with full DP subjects and similarly extend to possessor constructions, with differences in surface realization tracking the distinct prosodic structures associated with each configuration. Intervening focus markers and case markers are shown to systematically disrupt these processes as predicted by changes in linear adjacency. The same sensitivity to adjacency and prosodic structure is also observed with pronominal objects, further demonstrating that these processes are not confined to the subject-auxiliary boundary. Since the Kono paradigm is not suppletive, the empirical precondition for needing post-syntactic morphological operations is not met. The Kono data thus bear on the division of labor between phonology and morphology in accounting for STAMP paradigmatic variation cross-linguistically.

Keywords: morphophonology, adjacency, prosodic structure, portmanteaux, Mande, Kono

1 Introduction

STAMP patterns reflect surface configurations in which pronominal subjects and markers of tense, aspect, mood, and polarity appear to combine into single, often unsegmentable forms.¹ Such patterns are well-documented across multiple language families of the Macro-Sudan Belt, including Mande, Kwa, Gur, and Atlantic (Anderson 2011, 2015, 2016; Schreiber 2011; Konoshenko 2020; Garvin et al. 2021, To appear, a.o.). The phenomenon is illustrated by the Kono (Central Mande) constructions in (1) and (2).²

¹Coined by Anderson (2015), the STAMP acronym captures the morphosyntactic features typically implicated: Subject, Tense, Aspect, Mood, and Polarity. The observation that subject and TAM markers combine into portmanteau-like forms in West African languages predates this coinage; see Creissels 2005; Anderson 2011; Schreiber 2011, among others.

²Standard Leipzig glossing is used except for the following: EXCL = exclusive, INCL = inclusive, LV = light verb, NPST = non-past, OBJ = object case marker, P = postposition, PRT = particle.

In each pair, the sentences differ only in tense, yet the STAMP forms do not transparently decompose into a recognizable pronoun and TAM marker across these constructions. While the 1SG forms *ná~mbé* in (1) share a nasal onset segment, the 2SG forms *já~ée* in (2) notably do not appear to share any segments.

- (1) a. *ná* *kánánà-à* *tó*
 1SG.PST door-OBJ close
 ‘I closed the door.’
 b. *mbé* *kánánà-à* *tón-dà* *wá*
 1SG.NPST door-OBJ close-PRT FOC
 ‘I will close the door.’
- (2) a. *já* *kánánà-à* *tó*
 2SG.PST door-OBJ close
 ‘You closed the door.’
 b. *ée* *kánánà-à* *tón-dà* *wá*
 2SG.NPST door-OBJ close-PRT FOC
 ‘You will close the door.’

The opacity in the relationship between surface forms extends throughout the paradigm. Table 1 shows that while some person/number forms exhibit overlapping segments (e.g., 1DU *mɔ́á~mɔ́é~mɔ́ní*), others share fewer segments (e.g., 3SG *àà~èè~àní*) or no segments at all (e.g., 2SG *já~ée~íní*).³

Table 1: Partial STAMP paradigm in Kono

P/N	PST	NPST	PROG
1SG	<i>ná</i>	<i>mbé</i>	<i>ní</i>
2SG	<i>já</i>	<i>ée</i>	<i>íní</i>
3SG	<i>àà</i>	<i>èè</i>	<i>àní</i>
1DU	<i>mɔ́á</i>	<i>mɔ́é</i>	<i>mɔ́ní</i>
1INCL	<i>mɔ́à</i>	<i>mɔ́è</i>	<i>mɔ́ní</i>
1EXCL	<i>nàà</i>	<i>mbèé</i>	<i>níí</i>
2PL	<i>wá</i>	<i>wé</i>	<i>óní</i>
3PL	<i>àà</i>	<i>àmbè</i>	<i>àní</i>

Often characterized as an areal feature, STAMP paradigms across West Africa vary in the degree to which their surface forms are phonologically predictable. Garvin et al. (2021, To appear) show that within such paradigms, some forms are decomposable into underlying pronominal and TAM components via regular phonological processes, with some forms appearing to be genuinely suppletive.

While the term “STAMP” is clearly a descriptive label for a surface pattern, it raises a question concerning what part of the grammar derives such complexity. The apparent irregularity in such forms is compatible with two very different sources. On the one hand, it could reflect genuine morphological portmanteaux, in which single exponents realize features from multiple syntactic nodes. On the other, it could arise from independent underlying components that interact phonologically to yield surface complexity. Distinguishing between these possibilities is an empirical question, and one that

³The relevant conditioning factors extend beyond person/number and tense/aspect to include argument position, among other possible factors, as discussed in more detail in § 4.

requires systematic investigation of both the morphosyntactic structures underlying these forms and the morphophonological processes operative in the language.

This paper undertakes that investigation for Kono. Based on original fieldwork with a native speaker, I provide the first morphosyntactic and morphophonological analysis of Kono STAMP patterns.⁴ The primary claim I advance is that the Kono STAMP paradigm is phonologically derived and does not reflect suppletion. The surface forms in Table 1 result from regular, independently motivated phonological processes applying to distinct exponents. No dedicated morphological mechanisms are needed to realize these forms.

1.1 Theoretical stakes: what it means to call a (STAMP) paradigm suppletive

Within realizational theories of morphology like Distributed Morphology (DM), post-syntactic operations are available to restructure syntactic outputs before phonological material is inserted (Halle and Marantz 1993, 1994; Halle 1997; Embick and Noyer 2001, 2007). Since such operations expand the inventory of mechanisms available to the morphological component of the grammar, their invocation should be empirically motivated. Whether a given STAMP paradigm requires such mechanisms depends on a prior question: is the paradigm suppletive?

Suppletion represents the limiting case of paradigmatic irregularity, where the relationship between alternants is arbitrary in form despite being regular in meaning and function (Osthoff 1899; Mel'čuk 1976; Bobaljik 2012, 2015). Familiar examples include English *go~went* (past tense), *good~better* (comparative), and *be~am~is~are* (person/number agreement), where the alternating roots cannot be related by any independently motivated phonological process of the language.

The characterization of suppletion as phonologically arbitrary alternation entails an important methodological consequence: the claim that a paradigm is suppletive carries a burden of proof. To establish suppletion, one must demonstrate not merely that surface forms *appear* irregular, but that no motivated phonological derivation relates them. Equally, a phonological derivation is only as strong as the independent motivation for the processes it invokes: they must be attested elsewhere in the language and applicable on general grounds, *ceteris paribus*. A paradigm whose surface forms show considerable variation may nonetheless prove phonologically predictable once appropriate underlying representations are posited and independently motivated phonological processes are identified, as Garvin et al. (2021, To appear); Rolle (2022); Russell (2024) have demonstrated for various STAMP paradigms. If a language's phonology can account for the observed variation, positing listed suppletive allomorphs introduces stipulation that the grammar does not require. Conversely, if the phonological processes needed to derive a paradigm are not independently attested, or require unmotivated stipulations of their own, the case for a strict phonological derivation weakens, and appeal to suppletion or other *morphophonological* machinery grows more attractive.⁵ The appearance of surface irregularity is a necessary but not sufficient condition for a suppletion claim.

⁴Previous work on Kono's phonological system includes Manyeh (1983) and Foday-Ngongou (1985), which may describe varieties distinct from the one investigated here.

⁵It is not necessarily the case that if a phonological process needed to explain a paradigm is *not* independently attested in the language, then a phonological derivation is strictly out. It could just be that this is the only environment in which the conditions are right for the phonological rule(s) to apply (cf. cases of so-called *phonologically conditioned suppletive allomorphy* (see Carstairs 1990; Nevins 2011; Rolle and Bickmore 2022)). Of course, such an explanation grows more stipulative if *no* other connections can be made (e.g., to prosodic domains, prosodic word structure, etc.).

When suppletion *is* established, it has direct consequences for morphological theory. Within DM, the baseline mechanism for morphological realization is Late Insertion equipped with a principle of contextual allomorphy, whereby Vocabulary Items compete to insert phonological material at individual terminal nodes, with competition governed by the Elsewhere Principle (Halle and Marantz 1993, 1994). This mechanism is sufficient for suppletion involving a single terminal node (e.g., for realizing $\sqrt{\text{GO}}$ in *go~wen(t)*). But for suppletive paradigms in which a single surface exponent realizes features originating from *multiple* syntactic terminal nodes, the grammar must provide some means of associating a single exponent with features from multiple sources. In standard DM, Fusion combines terminal nodes into a single complex node prior to Vocabulary Insertion, creating a target for a portmanteau Vocabulary Item (Halle and Marantz 1993; Embick and Noyer 2001, 2007). Other DM-internal approaches, such as Spanning (Svenonius 2012; Merchant 2015), derive portmanteau effects by widening the domain that a single Vocabulary Item can target, while other proposals involving non-terminal insertion depart more fundamentally from the standard insertion mechanism.

STAMP forms are directly relevant here, since they appear to realize features from distinct nodes (subject φ -features and TAM/polarity). For instance, Felice 2021, 2022 pursues a Fusion-based analysis for Gã, where STAMP forms exhibit surface patterns comparable to those in Kono. As illustrated in (3), the 1sg subject form *í* in the simple past alternates with *míí* in the progressive; the latter appears to be a single exponent realizing both subject and aspect features. To account for such an alternation, Felice proposes that a post-linearization Fusion operation targets certain subject pronouns (D, in Spec,Infl) and inflectional heads (Infl) and fuses them into a single complex node, as illustrated by (4).⁶ The output is then targeted by a portmanteau Vocabulary Item.

(3) STAMP in Gã (Felice 2021: 847, (5, 7))

- a. *í ná gbèé-i éɲɔ*
1SG see dog-PL two
'I saw two dogs.'
- b. *míí na bo*
1SG.PROG see 2SG.ACC
'I am seeing you.'

(4) $[+1, -\text{PL}]_{\text{D}} \frown [\text{PROG}]_{\text{Infl}} \rightarrow [+1, -\text{PL}, \text{PROG}]$

Crucially, this analysis presupposes that the Gã paradigm contains suppletive forms, and Fusion is the mechanism Felice 2022 invokes *given* that conclusion.⁷ For any STAMP paradigm, theory-internal mechanisms like Fusion become relevant only if the paradigm is genuinely suppletive. The prior question is whether the surface forms can be derived from independently inserted exponents via regular phonological processes. If they can, then no additional machinery is required, and none should be posited. The following subsection previews how this paper addresses that question for Kono.

⁶While a Spanning approach can capture certain portmanteau complexes by relaxing the single-node constraint on insertion, Felice observes that standard Spanning cannot derive the relevant STAMP configuration, since the subject pronoun and the inflectional head stand in a Spec-Head relation rather than a head-complement relation, and thus do not form a contiguous span in the relevant sense.

⁷See also Russell 2024, which argues that all but a few Gã STAMP forms are fully predictable from regular phonological processes in the language. The exceptional ones noted in Russell 2024 are precisely those which Felice 2021, 2022 derives via Fusion.

1.2 The analytical question for Kono: Preview of the proposal

I argue that the Kono STAMP paradigm is phonologically derived and does not reflect suppletion. The surface forms in Table 1 result from regular phonological processes which apply to independently inserted exponents. Pronouns and auxiliaries remain distinct morphosyntactic objects throughout the derivation; no dedicated morphological mechanism is needed to derive the forms. Table 2 previews this decomposition. The underlying pronoun and auxiliary, shown to the left of the arrow in each cell, give rise to the attested surface forms via independently motivated phonological processes.

Table 2: Decomposed STAMP forms in Kono: UF → SF

p/n	φ -PST	φ -NPST	φ -PROG
1SG	Ń -a → n -á	Ń -be → m -bé	Ń -ni → ní
2SG	í -a → j -á	í -be → é -é	í -ni → í -ní
3SG	à -a → à -à	à -be → è -è	à -ni → à -nì
1DU	mǎ -a → mǎ -á	mǎ -be → mǎ -é	mǎ -ni → mǎ -ní
1INCL	mô -a → mó -à	mô -be → mó -è	mô -ni → mó -nì
1EXCL	Ń: -a → n -àà	Ń: -be → m -bèé	Ń: -ni → níí
2PL	ó -a → w -á	ó -be → w -é	ó -ni → ó -ní
3PL	àN -a → àn -à	àN -be → àm -bè	àN -ni → à -ní

This analysis presupposes that the elements composing STAMP forms in Kono involve a subject pronoun occupying an argument position and a separate auxiliary, rather than subject φ -agreement realized on a single functional head. In § 2, I establish the structural status of pronouns on independent grounds. There I show that the complementary distribution between pronouns and DP subjects, the behavior of hanging topic constructions, and preliminary intervention evidence from case markers all provide evidence that the Kono STAMP forms involve two distinct morphosyntactic objects: a subject pronoun and a separate auxiliary.

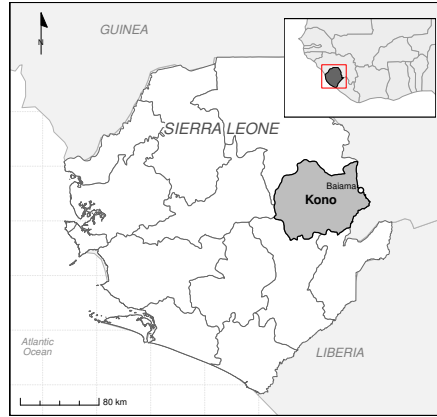
The paper then develops a cumulative case for the claim that these objects interact phonologically rather than being combined morphologically. § 3 establishes a systematic parallelism wherein the phonological processes deriving pronominal STAMP forms are also operative when full DP subjects occur with auxiliaries, with the specific realization of these processes dependent on the prosodic relationship between the subject and the auxiliary. Notably, this parallelism extends beyond the subject-auxiliary boundary to possessor constructions, where analogous phonological interactions occur at the possessor-possesum juncture (§ 3.4). § 4 extends the empirical domain to pronominal objects and non-structural case, showing that the same phonological processes operate in these configurations. Case marking additionally provides an intervention diagnostic: when case-marked internal arguments surface as derived subjects in unaccusative constructions, the case marker linearly intervenes between the subject and the auxiliary, systematically disrupting the phonological processes that would otherwise produce the surface forms in Table 1 while feeding other regular phonological processes. § 4.3 demonstrates that the same sensitivity to adjacency and prosodic structure is observed with pronominal objects, further extending the empirical domain of the pattern beyond subject-auxiliary and possessor constructions. That this sensitivity to linear adjacency holds across disparate morphosyntactic configurations is expected if phonological rules apply post-insertion to concatenated exponents, and it is at best unexplained if the morphology creates a single portmanteau node for deriving these forms.

2 Language background

2.1 Basic facts of Kono

Kono [ISO 639-3: kno] is a Central Mande language of the Niger-Congo phylum, spoken by approximately 350,000 speakers in and around the Kono District in the Eastern Province of Sierra Leone, as shown in Figure 1 (Eberhard et al. 2026). The data presented in this paper were collected by the author through elicitation sessions with a native speaker of Kono from Baiama, Kono District. Unless otherwise noted, all examples are from these sessions.

Figure 1: Map of Kono District, Eastern Province, Sierra Leone



The basic word order in Kono shows an S-Aux-O-V(X) pattern, where the auxiliary immediately follows the subject and the verb appears in clause-final position, sometimes followed by a particle and/or PP adjuncts. This configuration is broadly characteristic of Mande languages (Schreiber 2011; Konoshenko 2020). The pattern is illustrated in (5) with the progressive auxiliary and in (6) with the past auxiliary.

- (5) bòndú-ní sàni-ò tè-à,...
Bondu-PROG bottle-OBJ break-PRT
'Bondu was breaking the bottle,...'
- (6) bòndú-á sàni-ò tè
Bondu-PST bottle-OBJ break
'Bondu broke the bottle.'

The auxiliary and the preceding subject are phonologically interdependent, as reflected in the transcription. The evidence for this characterization and its consequences for the phonological processes discussed in this paper are developed in § 3.⁸

⁸Negation is a common component of STAMP patterns cross-linguistically (Anderson 2011, 2016; Garvin et al. 2021, To appear; Grimm 2025). Kono exhibits negative constructions in which the subject and a negative marker appear to combine in ways that parallel the affirmative STAMP patterns discussed in this paper. In (ii), for instance, the 1SG form is not clearly segmentable from the negative particle *ma*. Some negated clauses additionally show what appears to be a post-verbal particle that is segmentally identical to what is labeled as the progressive auxiliary /ni/ in this paper; it is not yet clear whether this particle and the affirmative progressive constitute the same morpheme. See also Foday-Ngongou 1985 for discussion of phonological aspects of negation in Kono. A full analysis of negative STAMP forms is beyond the scope of this paper and is left for future research.

The consonant inventory of Kono is given in (7). The inventory includes a doubly articulated labial-velar stop /g̠b/, which patterns distinctly from the simple voiced obstruents /b/ and /d/.

(7) /p b t d k g̠b N m n ŋ ɲ s f j w tʃ /

The symbol /N/ is used throughout this paper to represent an underlying nasal segment that is underspecified for place of articulation; it assimilates in place to a following consonant and surfaces as [n] by default (i.e., before vowels). This segment appears in several morphological contexts, including the plural suffix and certain pronominal forms, and plays a central role in deriving the surface forms discussed in §§ 3 and 4. Among the nasals in (7), /m/ and /ɲ/ are clearly contrastive, as supported by the near minimal pair in (8). The status of the velar nasal [ŋ] is less certain: it may be phonemic, but there are clear instances where its realization is best analyzed as an allophone of /N/, as in (9).

(8) a. mɛ̀ɛ̀ ‘knife’
b. ɲɛ̀ɛ̀ ‘fish’

(9) a. ɲ́-kámà̀nè ‘my shoulder’
b. ɲ́-mámà̀kái ‘my grandfather’

The vowel inventory, given in (10), includes seven oral vowels with an ATR distinction among mid vowels: /e/ and /o/ are [+ATR], while /ɛ/ and /ɔ/ are [−ATR]. This distinction is transcribed straightforwardly throughout the paper, with separate IPA symbols for each ATR pair. Kono also has a phonemic vowel length contrast; long vowels are transcribed as sequences of identical vowels (e.g. /aa/, /ee/). Nasalized vowels are found throughout Kono but are not phonemic.

(10) /i e ɛ a o ɔ u /

Kono is a tonal language with two level tones, high (H) and low (L), which can combine into contour tones on a single tone-bearing unit. Tone distinguishes lexical items, as illustrated by the minimal pairs in (11) and (12).

(11) a. fáá ‘swollen’
b. fàà ‘heart’

(12) a. tú ‘oil’
b. tù ‘(to) hit’

Tone also plays a role in the morphophonology of functional elements. The surface tones on auxiliaries in Table 2, for instance, generally track the tone of the preceding pronoun, though not uniformly: the progressive auxiliary /ni/ surfaces with a high tone after the low-toned 3PL pronoun, for example. A full analysis of the tonal system is not attempted here. Tonal patterns are analyzed where they bear directly on the segmental processes under investigation, but more complex tonal interactions are noted descriptively and left for future research (see also Manyeh 1983 and Foday-Ngongou 1985 for further discussion of Kono tonal patterns). Whether auxiliaries carry underlying tone and/or receive it via spreading from adjacent elements is largely orthogonal to the central claims of this paper, though the question is revisited briefly where relevant.

(i) n-á bwí dɔ̀ɔ̀ kúnú
1SG-PST run LV yesterday
‘I ran yesterday.’

(ii) máá bwí dɔ̀ɔ̀-ɲí
1SG.PST.NEG run LV-NI
‘I did not run.’

2.2 The Kono STAMP system

Two distinct questions arise for any STAMP paradigm. The first concerns the structural status of the “S” (subject) component: is it a pronoun occupying an argument position, or φ -feature agreement realized on the auxiliary? The second concerns exponence: are subject-referencing features and TAM features expounded by distinct Vocabulary Items, or bundled on a single node and realized by a portmanteau exponent? These questions are independent. An agreement analysis does not entail a single terminal, as [Rolle \(2022\)](#) demonstrates for Ebira, where STAMP morphs are analyzed as φ -agreement yet decomposed into sub-STAMP morphs with distinct underlying exponents. Conversely, a pronominal analysis does not necessarily require two terminals at the point of insertion, since post-syntactic Fusion could combine a pronoun and an auxiliary node into one target for insertion ([Felice 2021, 2022](#)). This section addresses the first question and establishes that Kono STAMP forms involve a genuine pronoun in an argument position. The second question is addressed in §§ 3 and 4.

The structural status of the “S” component is not predetermined by the STAMP label itself. [Anderson \(2015: 152\)](#) highlights the “ambiguous nature of STAMP morphs”, noting that such elements “appear to function simultaneously as pronominals and something like auxiliaries” and that different analysts have reached different conclusions for individual languages. The question must therefore be investigated on a language-by-language basis. In Ebira, for example, independent subject pronouns and STAMP morphs co-occur, thus supporting [Rolle’s \(2022\)](#) analysis that the “S” reflects φ -feature agreement with the structural subject.

In Kono, the distributional evidence points in the opposite direction: STAMP forms involve a genuine pronominal argument. To see this, consider a simple clause with a pronominal STAMP form, as in (13), where the 3PL pronoun *àN* is immediately left-adjacent to the progressive auxiliary *ní*.⁹

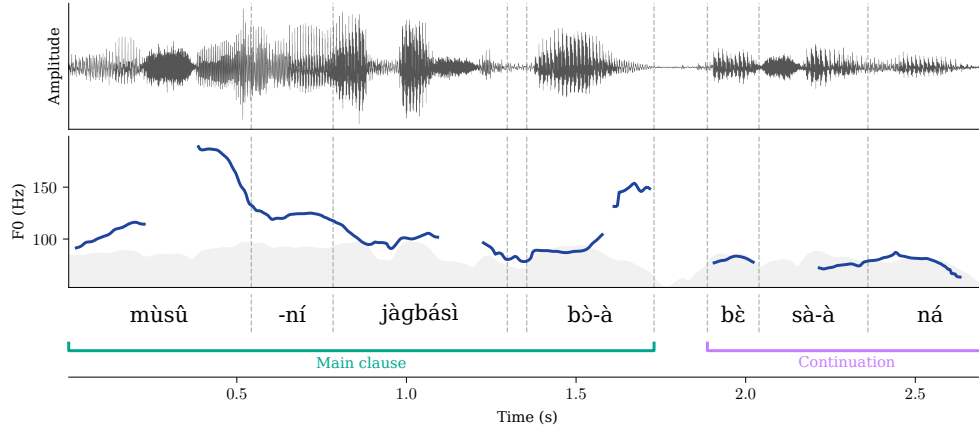
- (13) *à-ní jàgbàsì b̀-à, bè sà-à ná*
 3PL-PROG onion harvest-PRT then Sa-PST come
 ‘They were harvesting onions when Sa arrived.’

The primary evidence that the subject-referencing element in (13) is a pronoun occupying a structural argument position, rather than φ -feature agreement with a null subject, comes from complementary distribution. When a full DP occupies the subject position, no pronominal element appears; the subject DP precedes the auxiliary directly. This is illustrated in (14), where the DP *mùsû* ‘(the) women’ is followed immediately by the progressive auxiliary *ní*, and co-occurrence with a pronoun is unacceptable. Pronouns and full DPs thus compete for the same structural position—a hallmark of pronominal arguments, not agreement, since agreement morphology characteristically co-occurs with full DP subjects. [Figure 2](#) shows the prosodic contour of (14), where no prosodic break separates the subject and the auxiliary.

- (14) *mùsû (*-à) -ní jàgbàsì b̀-à, bè sà-à ná*
 woman.PL (*-3PL) -PROG onion harvest-PRT then Sa-PST come
 ‘The women were harvesting onions when Sa arrived.’

⁹The auxiliary in these examples is the progressive *ní*, which is also the auxiliary whose STAMP forms are most transparently segmentable across [Table 1](#): the onset /n/ surfaces in every person/number cell. The less transparent auxiliaries, whose interaction with subject pronouns gives rise to the more opaque-looking portions of the paradigm, are the focus of the morphophonological part of the analysis in §§ 3 and 4.

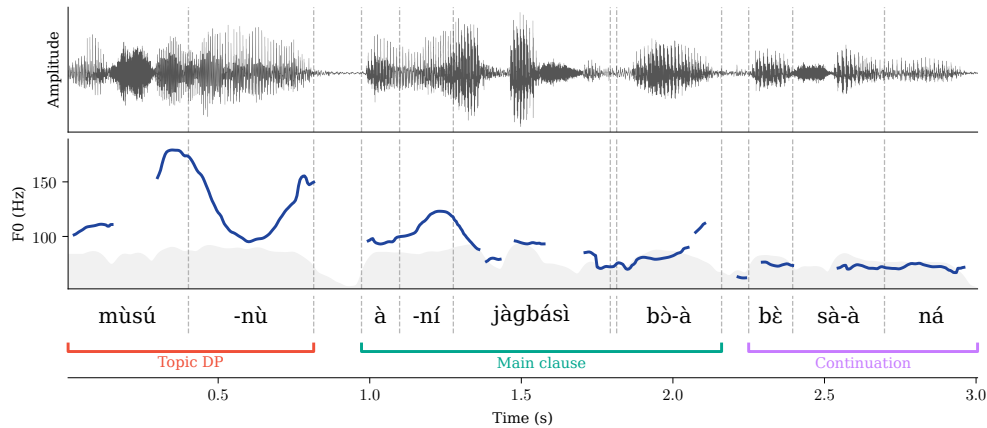
Figure 2: Prosodic contour of (14)



Further evidence comes from constructions in which a full DP and a pronoun both appear in the same clause, as in (15). In this construction, the 3PL pronoun *àn* serves as the structural subject, and the DP *mùsù-nù* ‘(the) women’ occupies a left-peripheral position (hanging topic), prosodically separated from the clause by a slight pause and rising intonation at its right edge, as illustrated in Figure 3. When the two co-occur, it is the pronoun, not the DP, that is left-adjacent to the auxiliary and participates in the phonological processes characteristic of STAMP forms.

- (15) mùsù-nù, à-ní jàgbàsì bò-à, bè sà-à ná
 woman-PL 3PL-PROG onion harvest-PRT then Sa-PST come
 ‘The women, they were harvesting onions when Sa arrived.’

Figure 3: Prosodic contour of (15), showing the prosodic boundary after the hanging topic DP



Coordinated expressions extend this generalization. In (16), the coordinated DP *demusu ni musu-nu* ‘child and women’ precedes the clause and is separated by the same prosodic break observed with the hanging topic in (15) (cf. Figure 3). The 3PL pronoun appears in the structural subject position, immediately left-adjacent to the auxiliary.

- (16) démúsù nì mùsù-nù, *(àm-bé) jàgbàsì bò-à wá
 child and woman-PL *(3PL-NPST) onion harvest-PRT FOC
 ‘The child and the women, they will harvest the onion.’

Notably, coordinated DP expressions do not occupy the structural subject position directly in Kono. Example (17) shows that the coordinated expression itself cannot appear immediately before the auxiliary. This pattern parallels the hanging topic construction in (15): in both cases, when a DP is external to the clause, a pronoun fills the structural subject position.

- (17) *[démúsù nì mùsú-m] -bé jàgbàsì bò-à wá
 child and woman-PL -NPST onion harvest-PRT FOC
 Intended: ‘The child and the woman will harvest the onion.’

A final piece of evidence comes from the behavior of case-marked arguments in unaccusative constructions. Certain Kono verbs assign non-structural (lexical) case to their internal arguments, realized as a suffix on the argument. In transitive clauses like (18), the case marker *-do* appears on the object *sàni-tfê-n* ‘these bottles’, while the DP subject *Bòndú* occupies the pre-auxiliary position.

- (18) bòndú -ní sàni-tfê-n-dó tè-à,...
 Bondu -PROG bottle-DET-PL-**OBJ** break-PRT
 ‘Bondu was breaking these bottles,...’

In the unaccusative variant in (19), the same case-marked internal argument surfaces in subject position, immediately preceding the auxiliary, and no additional subject-referencing element co-occurs. This is the same complementary distribution observed with subjects of transitives in (14): when a DP fills the structural subject position, the pronominal element is excluded. The pattern holds for pronominal internal arguments as well. In (20), the 3PL pronoun bears the case marker and occupies the subject position, and here, too, a second subject-referencing element is excluded.

- (19) sàni-tfê-n-dó (*à) -ní tè-à,...
 bottle-DET-PL-**OBJ** (*3PL) -PROG break-PRT
 ‘These bottles were breaking,...’
- (20) àn-dó (*à) -ní tè-à, bè...
 3PL-**OBJ** (*3PL) -PROG break-PRT, then
 ‘They were breaking, then...’

That the subject-referencing element in these constructions itself bears case morphology is also consistent with its status as an argument, since case is characteristically a property of arguments rather than agreement morphology. A more detailed analysis of the case-marking patterns and their consequences for the exponence question is developed in § 4.

The evidence presented in this section strongly suggests that Kono STAMP forms involve two morphosyntactic objects—a subject pronoun occupying an argument position and an auxiliary—rather than φ -agreement realized on a single functional head. The complementary distribution between pronouns and DP subjects, the obligatory appearance of a pronoun when DPs are external to the clause (as hanging topics), and the behavior of case-marked arguments in unaccusatives all point in the same direction. This structural generalization is a necessary precondition for the central analytical question of this paper, but it does not resolve that question on its own. As discussed in § 1, whether forms in a STAMP paradigm are suppletive or phonologically derived depends on the relationship between surface forms and the underlying exponents of the pronoun and the auxiliary. The following sections show that independently motivated phonological processes, applying to these now-established distinct objects, give rise to the attested forms.

3 Parallelism: DP and pronominal subjects

This section provides the primary evidence that the STAMP forms in Table 1 are phonologically derived rather than suppletive. The evidence comes from a systematic parallelism between the phonological processes observed at the boundary between a full DP subject and an auxiliary and those observed at the boundary between a subject pronoun and an auxiliary. This parallelism is expected if pronouns and auxiliaries are independently inserted exponents whose surface interaction is governed by the phonological grammar. If, on the other hand, pronominal STAMP forms were single, unanalyzable portmanteaux, there would be no principled reason for their surface realization to track the phonological conditioning observed with full DP subjects.

Before presenting this evidence, I first lay out the underlying representations and prosodic structures that the following sections develop. I propose Kono pronouns have the underlying forms represented in Table 3. The 1SG and 1EXCL pronouns are syllabic nasals, underspecified for place. 1SG /ŋ̃/ carries a high tone. The 1EXCL pronoun carries a low tone, but is further distinguished by an additional property: a vowel-lengthening effect consistently observed at the left edge of a following morpheme, with the low tone of the pronoun appearing on the lengthened portion. I represent this as /ŋ̃:/, treating the pronoun as underlyingly bimoraic, with the additional mora responsible for the observed lengthening. 1DU /m̌/ and 1INCL /m̌/ each carry a contour tone on a single tone-bearing unit. 2SG /í/ and 2PL /ó/ are high vowels. The 3SG pronoun is /à/. The 3PL pronoun /àN/ ends in a non-syllabic, underspecified nasal; while it is plausibly decomposable into /à/ (3SG) + /-N/ (PL), paralleling the plural suffix that appears on nouns, this paper treats the 3PL /àN/ as a unit throughout. In general, the tone of the pronoun spreads rightward to the following tone-bearing unit at morphosyntactic boundaries, accounting for the tonal patterns observed in Table 2 and throughout § 3. Exceptions to this pattern are noted where they arise.

Table 3: Pronoun inventory (UF)

	SG	DU	PL	
			INCL	EXCL
1	ŋ̃	m̌	m̌	ŋ̃:
2	í	—		ó
3	à	—		àN

The underlying forms proposed for the three auxiliaries are given in Table 4. PST /a/ is purely vocalic. NPST /be/ has an onset /b/ that surfaces only after nasals and is absent after vowels. PROG /ni/ has an onset /n/ that surfaces invariably across all environments. These characterizations are motivated by the data presented in the following subsections.¹⁰

Table 4: Auxiliary inventory (UF)

Aux	UF
PST	a
NPST	be
PROG	ni

¹⁰The underlying forms in Table 4 are given without tonal specification on the auxiliary itself; a full investigation of whether and how grammatical tone is distinguished from lexical tone is beyond the scope of this paper.

I propose that the parallelism between full DP subjects and pronominal subjects in STAMP constructions is mediated by prosodic structure. On the analysis developed here, the auxiliaries in Table 4 are enclitics, parsed into the phonological phrase containing the preceding subject (cf. Selkirk 1996 on the prosody of enclitics). The prosodic structure that results, however, depends on the prosodic status of the preceding element. Full DP subjects form their own prosodic words, while pronouns do not. This distinction follows from the typology of function-word prosodification developed in Selkirk 1996, on which “function words” may be incorporated into an adjacent prosodic word rather than projecting one independently.

For Kono, I propose the following two configurations. When an auxiliary combines with a full DP subject, the auxiliary is parsed as part of the phonological phrase but outside of the DP’s prosodic word boundary, as schematized in (21), where ω represents a prosodic word and ϕ a phonological phrase (Nespor and Vogel 1986). When an auxiliary combines with a pronoun, however, neither element *independently* projects a prosodic word; rather, the two are parsed together as a single prosodic word (itself contained within a phonological phrase), as in (22).¹¹

(21) (ϕ (ω DP) aux)

(22) (ϕ (ω pro aux))

In both configurations, the surface forms that arise at the subject-auxiliary boundary are predictable from independently motivated phonological processes, but the prosodic domain determines which processes apply and how. The following subsections present evidence for this analysis.

The remainder of this section is organized by auxiliary. For each of the auxiliaries identified in Table 4, I present data from full DP subjects alongside data from pronominal subjects, demonstrating that the surface forms in each case are predictable from general phonological processes, while also establishing the empirical case for the prosodic distinction proposed above. Where available, I also show that when a focus marker linearly intervenes between the subject and the auxiliary, the adjacency that gives rise to the relevant phonological process is disrupted, and the underlying forms of the pronoun and the auxiliary surface under distinct conditions due to the intervening material. Finally, § 3.4 extends the parallelism beyond the subject-auxiliary boundary, showing that analogous phonological interactions involving pronouns occur DP-internally at the possessor-possessum juncture.

3.1 Past auxiliary

The past auxiliary provides the most transparent starting point. The auxiliary surfaces invariably as [a], showing no segmental alternation regardless of the phonological properties of the preceding subject.¹² With the past auxiliary, any segmental phonological adjustment at the subject-auxiliary boundary is borne entirely by the final segment of the subject. With DP subjects, this transparency is immediately visible. Example (23) provides a baseline, where the subject *bondu* is followed directly by the auxiliary [a], with no interaction between the subject-final vowel and the auxiliary.

¹¹That the pronoun and the auxiliary are parsed into a single prosodic word despite not forming a syntactic constituent raises questions concerning the mapping between syntactic structure and prosodic domains. This question is revisited in light of additional evidence in § 4.3 and addressed briefly in § 5.

¹²The past auxiliary surfaces invariably as [a] in the transitive constructions discussed in this section. In certain other configurations, notably unaccusative constructions with case-marked subjects, the auxiliary is not overtly realized, a pattern discussed in § 4. The generalization stated here is restricted to the environments relevant to the parallelism argument.

- (23) bòndú -á káṇánà-à tó
 Bondu -PST door-OBJ close
 ‘Bondu closed the door.’

When the subject ends in a nasal, as with the plural suffix /-N/ in *musu-n* ‘women’ in (24), the nasal surfaces in its default realization [n] before vowels. The auxiliary itself remains [a].

- (24) mùsú-n -á jàgbási-à béjà
 woman-PL -PST onion-OBJ fall
 ‘The women dropped the onion.’

Pronominal subjects show the same pattern. In (25), the 3SG pronoun /à/, being vowel-final, precedes the auxiliary without segmental interaction. In (26), the 1SG pronoun /Ń/, an underspecified nasal, surfaces as [n] before the vowel-initial auxiliary—the same default coronal realization observed with the plural suffix on NP subjects. In (27), the 3PL pronoun /àn/ parallels both the plural NP and the 1SG pronoun: the final nasal surfaces as [n] before the vowel-initial auxiliary.

- (25) /a a/
 à -à káṇánà-à tó
 3SG -PST door-OBJ close
 ‘He/she closed the door.’

- (26) /Ń a/
 n -á káṇánà-à tó
 1SG -PST door-OBJ close
 ‘I closed the door.’

- (27) /àn a/
 àn -á jàgbási-à béjà
 3PL -PST onion-OBJ fall
 ‘They dropped the onion.’

The 2SG and 2PL pronouns exhibit one additional process at the pronoun-auxiliary boundary, namely prevocalic glide formation. In (28), the 2SG pronoun’s high front vowel /i/ surfaces as the palatal glide [j] before the auxiliary vowel, and in (29), the 2PL pronoun’s high back vowel /ó/ surfaces as the labial glide [w]. In both cases, the vowel hiatus created by the concatenation of the pronoun and the vowel-initial auxiliary is resolved by the high vowel surfacing as the corresponding glide (a cross-linguistically well-attested repair strategy, cf. Casali 1997).

- (28) /í a/
 j -á káṇánà-à tó
 2SG -PST door-OBJ close
 ‘You closed the door.’

- (29) /ó a/
 w -á dǐ-tǐè
 2PL -PST cry-LV
 ‘You all cried.’

Notably, prevocalic glide formation does not apply at the DP-auxiliary boundary; *bòndú* in (23) ends in a high vowel but does not trigger glide formation before /a/ (e.g. **bòndw-á*).¹³ This asymmetry is consistent with the prosodic distinction proposed in (21) and (22): within ω , vowel hiatus is resolved; across an ω boundary, it is tolerated. Further conditions on prevocalic glide formation emerge in § 3.2, where the non-past auxiliary presents a more complex set of environments.

Further evidence for the decomposability of the subject and the auxiliary comes from constructions involving focus marking. Kono marks narrow focus with a morpheme that attaches to the focused constituent. When the focused element is the subject, the focus marker linearly intervenes between the subject and the auxiliary, disrupting the adjacency that conditions the phonological processes described above. The examples that follow involve answers to subject-focused questions (e.g., *Who closed the door?*), with the focused element indicated by italicized caps in the translation.

The focus marker shows a three-way surface alternation of its onset segment, *-wáN ~ -fáN ~ -áN*. These alternations point to an underlying form /wáN/.¹⁴ The initial /w/ of the focus marker is retained after vowel-final pronouns, surfacing as [w], as illustrated in (30).

- (30) /à wáN a/
à -**wán** -à káṇánà-à tó
3SG -FOC -PST door-OBJ close
‘HE/SHE closed the door.’

The focused 2SG and 2PL pronouns in (31) and (32) provide a direct test of the adjacency conditions on prevocalic glide formation established above. When the focus marker intervenes between the pronoun and the auxiliary, the high vowels /í/ and /ó/ surface faithfully as [í] and [ó], respectively, rather than as glides (cf. (28) and (29)). The high vowels surface as glides only when immediately adjacent to the auxiliary vowel. The focus marker removes this adjacency, thereby bleeding prevocalic glide formation.

- (31) /í wáN a/
í -**wán** -à káṇánà-à tó kúnú
2SG -FOC -PST door-OBJ close yesterday
‘YOU(sg) closed the door yesterday.’
- (32) /ó wáN a /
ó -**wán** -à káṇánà-à tó kúnú
2PL -FOC -PST door-OBJ close yesterday
‘YOU(pl) closed the door yesterday.’

The focused 1SG and 3PL pronouns in (33) and (34) illustrate distinct phonological alternations at this boundary, namely place assimilation of the nasal. In both examples, the initial /w/ of the focus marker undergoes fortition after a nasal, surfacing as [f]. In turn, the final nasal of the pronoun surfaces as [ŋ], assimilating in place to labiodental [f]. The nasal thus assimilates to the output of fortition, not to the underlying /w/.

¹³Further evidence for this domain restriction, from environments that control for syllable structure, is presented in § 3.4.

¹⁴The final nasal /N/ of the focus marker behaves like other underspecified nasals in the language, surfacing in its default coronal realization [n] before vowels and assimilating in place before consonants.

(33) /ń wáN a/
 ń -fán -à káńánà-à tó kúnú
 1SG -FOC -PST door-OBJ close yesterday
 ‘I closed the door yesterday.’

(34) /àN wáN a/
 à ń -fán -à káńánà-à tó kúnú
 3PL -FOC -PST door-OBJ close yesterday
 ‘THEY closed the door yesterday.’

Comparing (33) and (34) with the non-focused forms in (26) and (27), where the same nasals surface as [n] before the vowel-initial auxiliary, we see that the nasal is sensitive to the segment that immediately follows it. When no focus marker is present, the nasal surfaces in its default coronal realization before the auxiliary vowel /a/. When the focus marker intervenes, the following segment is [f], and the nasal assimilates accordingly. This process of place assimilation will prove central to the analysis of the non-past auxiliary in § 3.2.

After vowel-final DP subjects, the focus marker’s initial /w/ deletes and the focus marker surfaces as [-án], as in (35) (cf. *Sà-wán-à). The contrast between [-án] after Sà in (35) and [-wáN] after 3SG in (30) is particularly revealing. Both subjects end in the same low vowel, yet they condition different surface forms of the focus marker: /w/ surfaces after the pronoun but is absent after the DP subject.

(35) sà -án -à dìí-tjè kúnú
 Sa -FOC -PST cry-LV yesterday
 ‘SA cried yesterday.’

This pattern tracks the prosodic distinction between DP and pronominal subjects (cf. (21) and (22)). To account for this asymmetry, I propose that the focus marker is parsed into the prosodic word with the pronoun and the auxiliary, as schematized in (36); within this domain, the onset /w/ surfaces faithfully. After a full DP subject, the focus marker falls outside the DP’s prosodic word, and /w/ is absent, as in (37).

(36) (ω à wáN a)
 à -wán -à káńánà-à tó
 3SG -FOC -PST door-OBJ close
 ‘HE/SHE closed the door.’

(37) (ω sà) wáN a
 sà -án -à dìí-tjè kúnú
 Sa -FOC -PST cry-LV yesterday
 ‘SA cried yesterday.’

The focus marker’s own onset alternation [wáN]~[án] thus independently tracks the prosodic distinction between pronouns and full DP subjects proposed above.¹⁵ The focus marker data demonstrate that the phonological processes observed at the subject-auxiliary boundary are sensitive to linear adjacency and disrupted in predictable ways when additional morphological material intervenes.

¹⁵ Alternation in the onset of the focus marker is also observed when it appears on constituents other than subjects, such as the focused objects in (i) and (ii). Such examples further illustrate that the conditioning factors for the focus marker’s initial segment are indeed phonological/prosodic rather than specific to morphosyntactic position/identity.

3.2 Non-past auxiliary

The non-past auxiliary presents a richer set of phonological interactions than the past auxiliary, but the same core principle holds: the surface alternations at the subject-auxiliary boundary are predictable from the phonological properties of the adjacent segments and hold similarly across both full DP subjects and pronominal subjects, *mutatis mutandis*.

With nasal-final subjects, the auxiliary /be/ surfaces as [be], and the preceding nasal undergoes place assimilation to the bilabial onset. In (38), the plural DP subject *mùsú-N* ‘women’ precedes the non-past auxiliary. The plural suffix /-N/ assimilates to [m] before the bilabial /b/ of the auxiliary, and the auxiliary surfaces as [be] with its onset intact.

- (38) /mùsú-N be/
 mùsú-m -bé jàgbàsì-à béjà-à
 woman-PL -NPST onion-OBJ fall-PRT
 ‘The women are dropping the onion.’

Nasal-final pronouns like 1SG and 3PL exhibit exactly this pattern. In (39), the 1SG /ŋ/ assimilates to [m] before the auxiliary’s onset, and the auxiliary surfaces as [be]. Likewise with 3PL /àN/ in (40).

- (39) /ŋ be/
 ń -bé kánánà-à tón-dà wá
 1SG -NPST door-OBJ close-PRT FOC
 ‘I will close the door.’

- (40) /àN be/
 àm -bè kánánà-à tón-dà wá
 3PL -NPST door-OBJ close-PRT FOC
 ‘They will close the door.’

These forms are directly parallel to the plural DP in (38). In each case, the final nasal of the subject assimilates in place to the following bilabial, and the auxiliary surfaces with its onset intact. As with the past auxiliary (§ 3.1), the nasal’s surface form is determined by the phonological features of the immediately following segment, regardless of the morphosyntactic identity of the subject.

We now turn to vowel-final subjects, where the auxiliary’s onset is systematically absent. In (41) with the DP subject *Sà*, the auxiliary’s onset /b/ does not surface, and the auxiliary vowel surfaces as [ɛ] rather than its underlying [e].

- (41) /sà be/
 sà -è tǽnà-à fían-dà
 Sa -NPST house-OBJ sweep-PRT
 ‘Sa will sweep the house.’

The lowering of the auxiliary vowel reflects vowel assimilation in the context of a preceding vowel. While the assimilation is unidirectional in (41), we do observe cases where the assimilation appears

(i) bòndú-á sàni-tǽ-án-dò tè
 Bondu-PST bottle-DET-FOC-OBJ break
 ‘Bondu broke *THIS* bottle.’

(ii) bòndú-á sàni-tǽ-ɲ-fán-dò tè
 Bondu-PST bottle-DET-PL-FOC-OBJ break
 ‘Bondu broke *THESE* bottles.’

bi-directional. In (42), both the final vowel of the DP subject *mànsá* and the auxiliary vowel surface as [ɛ], a quality intermediate between the final underlying /a/ in *mànsá* and the underlying /e/ of the auxiliary.¹⁶

- (42) /mànsá be/
 mànsé -è tǝ́nà-à fián-dà wá́ sínà
 chief -NPST house-OBJ sweep-PRT FOC tomorrow
 ‘The chief will sweep the house tomorrow.’

The 3SG pronoun /à/ in (43) shows a closely parallel pattern. Like in (42), the auxiliary’s onset is again absent, and the adjacent vowels interact. Both the pronoun vowel and the auxiliary vowel surface as [ɛ]. Two timing slots are preserved; the surface form [è-è] retains the segmental positions of both the pronoun and the auxiliary. Notably, the auxiliary’s onset alternation with both full DP subjects and pronominal subjects is identical: /b/ does not surface after a vowel in either configuration.

- (43) /à be/
 è -è káŋánà-à tòn-dà wá́
 3SG -NPST door-OBJ close-PRT FOC
 ‘He/she will close the door.’

Importantly, the absence of /b/ after vowels at the subject-auxiliary boundary is an instance of a broader restriction on intervocalic voiced obstruents in Kono. For instance, a similar alternation is observed outside the subject-auxiliary boundary, namely with verbal suffixes. In Kono, a post-verbal particle surfaces as [-da] when the verb stem ends in a nasal, and [-a] when the verb stem ends in a vowel.¹⁷ In (44), the nasal-final verb *fíáN* ‘sweep’ preserves the particle’s onset, and the stem-final nasal assimilates to [n] before /d/. In (45) with the vowel-final verb *tè* ‘break’, the particle surfaces without its onset. The voiced obstruent onset of the particle thus exhibits the same sensitivity to the preceding segment as the onset of the non-past auxiliary.¹⁸

¹⁶ The non-past auxiliary does not always surface segmentally after full DP subjects. After subjects ending in a high vowel, such as *Bondu*, the auxiliary is notably absent, as shown in (i). The generalization appears to be sensitive to the final segment of the subject rather than to its prosodic weight or syllable count. When an intervening focus marker provides a consonant environment for the auxiliary’s onset, the auxiliary surfaces overtly, as in (ii). A full account of this pattern is deferred to future work. The crucial observation for present purposes is that where the auxiliary does surface, its form is predictable from the phonological properties of the immediately preceding segment.

- | | |
|--|---|
| (i) bòndú tǝ́nà-à fián-dà wá́ sínà
Bondu house-OBJ sweep-PRT FOC tomorrow
‘Bondu will sweep the house tomorrow.’ | (ii) bòndú-ám-bè tǝ́nà-à fián-dà
Bondu-FOC-NPST house-OBJ sweep-PRT
‘BONDU will sweep the house.’ |
|--|---|

¹⁷ The particle’s distribution is sensitive to argument structure. It appears primarily with unaccusatives, but also occurs in transitive constructions with certain aspectual properties. Its morphosyntactic and semantic properties are not at issue here. The relevant observation is that its onset exhibits the same kind of alternation observed with the non-past auxiliary.

¹⁸ The restriction is not limited to morpheme boundaries. Word-internally, voiced obstruents are attested only after nasals (e.g., *bándà* ‘cotton’, *dumbü* ‘orange’, *sàmbà* ‘drum’, *kúndú* ‘short’), while voiceless obstruents appear freely in intervocalic position (e.g., *fétù* ‘pineapple’, *pùtù* ‘pepper’). The few apparent exceptions involve loanwords (e.g., *tébù* ‘table’ from English).

- (45) /tè da/
rín-bé tàà-ò tè -à
1SG-NPST calabash-OBJ break -PRT
'I am breaking the calabash.'

This pattern extends beyond the environments examined here. In § 4.1, the case marker *-do* is shown to exhibit the same pattern. Apparent exceptions, including the behavior of word-initial voiced obstruents in possessor constructions, are addressed in § 3.4.

Returning to the pronoun inventory with the non-past auxiliary, consider now the high-vowel pronouns 2PL /ó/ and 2SG /í/. Here we might expect glide formation to apply uniformly, as it does with the past auxiliary, yet an instructive asymmetry emerges. Example (46) shows that with the 2PL pronoun, glide formation does apply: the auxiliary's onset is absent, as expected after a vowel, and the pronoun's high back vowel /ó/ surfaces as the labial glide [w] before the auxiliary vowel [e], similar to how it surfaces as [w] before the past auxiliary (c.f., [w-á] in (29)).

- (46) /ó be/
w -é bėjà-à
2PL -NPST fall-PRT
'You all are falling.'

The 2SG pronoun, however, does not undergo prevocalic glide formation. In (47), the underlying 2SG-NPST sequence /i-be/ surfaces as [é-é]. While the auxiliary's /b/ onset is absent as expected, the pronoun remains vocalic. Rather than surfacing as the expected palatal glide [j], the pronoun vowel fully assimilates to the quality of the following auxiliary vowel, and both surface as [é] (cf. 2SG-PST [j-á] in (28)).

- (47) /i be/
 é -é kánjànà-à tòn-dà wá
 2SG -NPST door-OBJ close-PRT FOC
 ‘You will close the door.’

Note that this pattern differs from the vowel assimilation observed with the 3SG pronoun in (43), where both vowels converge on an intermediate quality [è-è]. Here, only the pronoun vowel adjusts; the auxiliary vowel surfaces faithfully. The difference in directionality plausibly reflects the featural distance between the input vowels: /a/ and /e/ are distant in both height and backness, motivating mutual accommodation, whereas /i/ and /e/ are both front vowels differing only in height, allowing straightforward assimilation of V_1 to V_2 . In both cases, however, two timing slots are preserved and the surface quality is derivable from the underlying segments.

The contrast between the two high vowels /o/ and /i/ in (46) and (47), respectively, follows from a restriction on prevocalic glide formation, whereby the process does not apply when the resulting glide would be homorganic with the following vowel. This type of restriction is well attested cross-linguistically (see [Casali 1997](#) on Niger-Congo) and is consistent with versions of the Obligatory Contour

Principle, which disfavors adjacent identical specifications (Leben 1973; Goldsmith 1976; Leben 1978; McCarthy 1979, 1986).

In (47), the 2SG's high front vowel /i/ and the auxiliary vowel /e/ share the feature [−back]. Because the palatal glide [j] is featurally identical to /i/ regarding place, devocalization of /i/ to [j] before [e] would yield a homorganic sequence ([j-e]) that retains the adjacent [−back] features. Under these conditions, glide formation is blocked, and the sequence undergoes assimilation instead, yielding [e-e]. In (46), on the other hand, the 2PL's vowel /o/ ([+back]) and the auxiliary /e/ ([−back]) differ in major place features. The resulting sequence [w-e] is heterorganic and thus well-formed.¹⁹

Prevocalic glide formation in a V_1V_2 sequence in Kono thus requires two conditions to be met simultaneously. First, V_1 must be a high vowel capable of devocalization (e.g., /i/ → [j], /o/ → [w]); the low vowel /a/ lacks a corresponding glide in the inventory and therefore cannot undergo this process (cf. the 3SG pronoun in (43)). Second, the two adjacent vowels must differ in major place features (such as [±back]), avoiding (the creation of) a homorganic sequence. Additionally, these two segmental conditions hold alongside a domain restriction: prevocalic glide formation applies within a single prosodic word but not across a prosodic word boundary (as observed in § 3.1; additional evidence for this domain restriction is presented in § 3.4). These conditions are schematized informally in (48).

(48) **Prevocalic Glide Formation** applies to V_1 in a sequence V_1V_2 iff:

1. V_1 is [+high]
2. V_1 and V_2 differ in [±back]

Domain: V_1 and V_2 must be contained within a single prosodic word (ω)

The past auxiliary, being [+back], satisfies the second condition with both high vowels, which accounts for why both 2SG and 2PL undergo prevocalic glide formation before /a/. The non-past auxiliary vowel /e/, being [−back, −round], satisfies it only with the back vowel /o/, not with the front vowel /i/. Where prevocalic glide formation is blocked, vowel assimilation applies instead.

As with the past auxiliary in § 3.1, focus marking on the subject disrupts the adjacency between the subject and the auxiliary, with predictable consequences for the processes just established. In (49), the focused 3SG pronoun surfaces as [à], not as [è]. The bidirectional vowel assimilation observed in (43) does not apply, since the pronoun vowel is no longer adjacent to the auxiliary vowel. Similarly, in (50), the focused 2SG pronoun surfaces as [í], not as [é]; neither vowel assimilation nor prevocalic glide formation applies, again because the adjacency that conditions these processes is disrupted (cf. (47)).

(49) /à wáN be/
à -wám -bè dìí-tfà-à sínà
3SG -FOC -NPST cry-LV-PRT tomorrow
‘HE will cry tomorrow.’

(50) /í wáN be/
í -wám -bè dìí-tfà-à sínà
2SG -FOC -NPST cry-LV-PRT tomorrow
‘YOU(sg) will cry tomorrow.’

¹⁹The sequence [je] is not phonotactically illicit in Kono (cf. the preposition *jě* ‘to’). The restriction observed here is specifically an active blocking of prevocalic glide *formation* as a repair for resolving hiatus. The process is blocked from creating a newly derived homorganic glide-vowel sequence, even if underlying sequences of that type are otherwise attested in the language.

At the focus marker-auxiliary boundary, the patterns reflect what is expected when a nasal precedes the non-past auxiliary, as established at the beginning of § 3.2. Since a consonant, not a vowel, now immediately precedes the auxiliary, the auxiliary's /b/ onset is retained, and the focus marker's final nasal /N/ assimilates to [m].

The generalization here is that the phonological shape of the auxiliary is determined by whatever segment is immediately adjacent to it in the linearized string, regardless of the morphosyntactic identity of the subject. The conclusion drawn from the past auxiliary in § 3.1 thus extends directly to the richer set of processes conditioned by the NPST auxiliary. Where PST showed a phonologically inert auxiliary and variation only on the subject side of the boundary, NPST shows variation on both sides.

3.3 Progressive auxiliary

The progressive auxiliary provides a useful contrast with the non-past auxiliary.²⁰ The underlying form is /ni/, and the nasal onset /n/ surfaces invariably, regardless of the phonological properties of the preceding segment. This is consistent with the onset alternation observed with NPST /be/ being a consequence of the intervocalic voiced obstruent restriction, not a general process at the subject-auxiliary boundary.

With DP subjects, the progressive auxiliary surfaces unchanged. In (51), the vowel-final subject *bòndú* is followed directly by [ní], with no segmental interaction between the final vowel of the subject and the onset of the auxiliary.

- (51) *bòndú -ní sà-n-dó tè-à kúnú, bè sà-à ná*
 Bondu -PROG bottle-PL-OBJ break-PRT yesterday, then Sa-PST come
 'Bondu was breaking the bottles, then Sa came.'

The form of the auxiliary is equally consistent when preceded by pronominal subjects. In (52), the 3SG pronoun /à/ precedes the auxiliary without segmental interaction, yielding two clearly distinct syllables [à-ní]. After 3PL /àN/, too, the auxiliary exhibits no segmental alternation, as shown in (53). With the 3PL pronoun, however, we do observe some phonological interaction. The final nasal /N/ of the pronoun leaves no detectable segmental trace before the auxiliary's onset. The surface form shows only the pronoun's vowel [à] followed by the auxiliary [ní], contrasting with its behavior before PST and NPST. Before the progressive, the non-syllabic nasal of 3PL /àN/ coalesces with the nasal onset of the auxiliary, leaving no segmental reflex.

- (52) /à ni/
 à -ní káǵánà-à tón-dà, mbe bondu na
 3SG -PROG door-OBJ close-PRT, then Bondu come
 'He was closing the door, then Bondu came.'

- (53) /àN ni/
 à -ní káǵánà-à tón-dà, mbe bondu na
 3PL -PROG door-OBJ close-PRT, then Bondu come
 'They were closing the door, then Bondu came.'

²⁰The label "progressive" reflects the auxiliary's most frequent use in the data presented here, where it typically conveys an ongoing or in-progress situation in the past. However, this auxiliary also appears in other constructions, including negative polarity and irrealis contexts. A more complete semantic characterization is beyond the present scope; I gloss it as PROG throughout for consistency.

As a consequence of coalescence, the 3PL-PROG form in (53) exhibits a sequence that is segmentally identical to that in (52). The two constructions, however, differ tonally: after 3SG, the auxiliary surfaces with a low tone, consistent with the general pattern of rightward tone spreading with pronouns, while after 3PL, the auxiliary surfaces with a high tone, despite the pronoun vowel also bearing a low tone. The 3PL pronoun's tone does not associate with the auxiliary; the pronoun's non-syllabic nasal /N/ is non-moraic and non-tone-bearing, and its coalescence with the auxiliary onset is purely segmental.

The 1SG and 1EXCL pronouns illustrate a useful point of contrast, as they show a different tonal pattern with coalescence. Underlyingly, both pronouns are underspecified nasals with distinct tonal specifications: 1SG /Ń/ bears high tone; 1EXCL /Ñ:/ bears a low tone and is associated with two moras. When these pronouns are adjacent to the progressive auxiliary, the pronoun nasal and the auxiliary onset coalesce, as illustrated in (54) and (55).

- (54) /Ń ni/
 Ø -ní káŋánà-à tòn-dà, mbe bondu na
 1SG -PROG door-OBJ close-PRT, then Bondu come
 'I was closing the door, then Bondu came.'

- (55) /Ñ: ni/
 Ø -nii káŋánà-à tòn-dà, mbe bondu na
 1EXCL -PROG door-OBJ close-PRT, then Bondu come
 'We(excl) were closing the door, then Bondu came.'

In the 1SG case, the auxiliary vowel surfaces with a high tone [ní], reflecting the pronoun's tonal specification. In the 1EXCL case, however, the auxiliary vowel surfaces with additional length and an initial low tone [nii], reflecting the bimoraic specification of the pronoun.²¹

Focus marking provides further evidence of this pattern. When the focus marker intervenes, coalescence is blocked and the pronoun nasal surfaces independently. In (56), the 1SG nasal retains its high tone. In (57), the 1EXCL nasal bears its low tone, and the lengthening effect appears on the focus marker vowel rather than on the auxiliary, consistent with the pattern observed in (55). In both cases, the surface forms are predictable from the independently established properties of the pronouns, the focus marker, and the auxiliary.

- (56) /Ń wáN ni/
 ní -fá -nì sàni-ò tè-à kúnú
 1SG -FOC -PROG bottle-OBJ break-PRT yesterday
 'I was breaking the bottle yesterday.'

²¹This lengthening effect is not specific to the progressive; it is observed consistently at all morphosyntactic boundaries following the 1EXCL pronoun, including with the other auxiliaries, as shown in (i) and (ii), and in possessor constructions (see § 3.4).

(i) m-bèé tǽnà-à fián-dà sínà
 1EXCL-NPST house-OBJ sweep-PRT tomorrow
 'We(excl) will sweep the house.'

(ii) n-àà swèè dào kúnú
 1EXCL-PST meat eat yesterday
 'We(excl) ate meat yesterday.'

- (57) /Ñ: wáN ni/
 ñ -fàá -ní sàni-ò tè-à kúnú
 1EXCL -FOC -PROG bottle-OBJ break-PRT yesterday
 ‘WE(excl) were breaking the bottle yesterday.’

The progressive auxiliary contributes several things to the analysis. Its invariant nasal onset provides a control case: the onset alternation of NPST is specific to voiced obstruents and does not extend to the nasal onset of PROG. The nasal coalescence patterns and tonal contrasts between 1SG and 1EXCL, and between 3SG and 3PL, show that the surface forms are derivable from the independently established properties of the pronouns and the auxiliaries. Focus marking supports the adjacency dependence of coalescence, consistent with the patterns established in §§ 3.1 and 3.2.

3.4 Extension: possessor constructions

The phonological processes identified at the subject-auxiliary boundary are not confined to the STAMP domain. Kono possessor constructions provide independent evidence both for the underlying forms proposed in Table 3 and for the prosodic domain distinctions developed above.

Kono morphosyntactically distinguishes inalienable possession from alienable possession. In inalienable possession, the possessor directly precedes the possessum with no intervening morphology, as shown in (58), while in alienable possession, a possessive morpheme appears between the possessor and possessum, as in (59)—a pattern well attested cross-linguistically and in Mande languages specifically (Nichols 1988; Innes 1967; Welmers 1976).²²

- | | | | |
|------|---|------|---|
| (58) | déndzè dè
child mother
‘the child’s mother’ | (59) | déndzè-à dáà
child-POSS pot
‘the child’s pot’ |
|------|---|------|---|

With pronominal possessors, the same contrast holds: in inalienable (60), the 1SG pronoun directly precedes the possessum, while in alienable (61), the possessive morpheme intervenes. The morphosyntactic distinction between inalienable and alienable possession is not the object of analysis here. Rather, the relevant observations concern the phonological behavior of the pronouns at the possessor boundary.

- | | | | |
|------|-------------------------------------|------|-------------------------------------|
| (60) | ń- dé
1SG- mother
‘my mother’ | (61) | n-á dáà
1SG-POSS pot
‘my pot’ |
|------|-------------------------------------|------|-------------------------------------|

The prosodic structures proposed for these two configurations follow from the property established at the beginning of § 3: pronouns do not form independent prosodic words, and the resulting prosodic parse depends on what adjacent material is available. In inalienable possession like in (60), no functional morpheme intervenes between the pronoun and the possessum. The possessum forms its own prosodic word, and the pronoun is parsed into the phonological phrase but outside the prosodic word, as schematized in (62). In alienable possession like in (61), the possessive morpheme *-a* provides a functional host, and the pronoun is parsed into a prosodic word with it, as in (63).

- | | | |
|------|---|-------------------------------|
| (62) | (_φ pro (_ω possessum)) | <i>Inalienable possession</i> |
| (63) | (_φ (_ω pro POSS) (_ω possessum)) | <i>Alienable possession</i> |

²²For tonal patterns in possessor constructions in varieties of Kono, see Manyeh 1983 and Foday-Ngongou 1985.

With pronominal possessors, the inalienable configuration in (62) thus mirrors the prosodic structure for the DP-aux configuration represented in (21), repeated below as (64): within the phonological phrase, an ω boundary separates a full DP from additional material that is not parsed into a prosodic word. The alienable configuration is more directly parallel to the pro-aux configuration in (22), repeated below as (65): the pronoun and an adjacent functional morpheme are parsed into a single prosodic word.

(64) $(\phi (\omega \text{ DP }) \text{ aux })$ *DP subject*

(65) $(\phi (\omega \text{ pro } \text{ aux }))$ *Pronominal subject*

Place assimilation of the underspecified nasal operates in the possessor domain as expected. In inalienable constructions, the pronoun's nasal surfaces with the place features of the following possessum onset—bilabial before /m/, velar before /k/, and so on—extending the range of observed place targets beyond those encountered at the subject-auxiliary boundary (Table 5). In alienable constructions, the nasal-final pronouns surface with the default coronal realization [n] before the possessive /-a/, paralleling the pattern observed before PST (Table 6).

Table 5: Inalienable possession

p/n	ϕ -grandfather	ϕ -shoulder	ϕ -hand	ϕ -brain
1SG	m-mámàkài	ŋ-kámànè	m-bóò	n-dʒîi
2SG	í-mámàkài	í-kámànè	í-bóò	í-îi
3SG	à-mámàkài	à-kámànè	à-bóò	à-îi
1DU	mò-mámàkài	mò-kámànè	mò-bóó	mò-îi
1INCL	mó-mámàkài	mó-kámànè	mó-bóò	mó-îi
1EXCL	m-màámàkài	ŋ-kààmànè	m-bóóó	n-dʒîi
2PL	ó-mámàkài	ó-kámànè	ó-bóò	ó-îi
3PL	àm-mámàkài	àŋ-kámànè	àm-bóò	àn-dʒîi

Table 6: Alienable possession

p/n	ϕ -POSS-banana	ϕ -POSS-medicine	ϕ -POSS-farm	ϕ -POSS-water
1SG	n-á máà	n-á bwîi	n-á bú	n-á íí
2SG	j-á máà	j-á bwîi	j-á bú	j-á íí
3SG	à-à máà	à-à bwîi	à-à bú	à-à íí
1DU	mò-á màà	mò-á bwîi	mò-á bú	mò-á íí
1INCL	mó-à máà	mó-à bwîi	mó-à bú	mó-à íí
1EXCL	n-à máà	n-à bwîi	n-àà bú	n-àà íí
2PL	w-á máà	w-á bwîi	w-á bú	w-á íí
3PL	àn-á máà	àn-á bwîi	àn-á bú	àn-á íí

Prevocalic glide formation provides more informative evidence. In the STAMP domain, this process is restricted to within a single prosodic word: it applies between a pronoun and an auxiliary parsed into the same ω but not across an ω boundary (§§ 3.1 and 3.2). The possessor domain offers an independent test of this restriction, where the pronoun vowel is a bare nucleus in both configurations, controlling for the syllabic-structural differences that complicate the STAMP evidence.

The 2PL pronoun /ó/ is diagnostic. In inalienable (66), the segmental conditions for prevocalic glide formation are met: /o/ is a high vowel capable of devocalization, and the following vowel /i/ differs in major place features. If glide formation occurred across a prosodic word boundary, we would expect /o/ → [w]. Yet the pronoun surfaces faithfully as [ó], with vowel hiatus tolerated. This is consistent with the ω boundary separating the pronoun from the possessum as in (62). In alienable (67), the same pronoun precedes the possessive morpheme /-a/, but here it surfaces as a glide, just as it does before PST /a/ (cf. (29)). This is expected if the pronoun and the possessive morpheme are parsed into a single ω , as in (63).

(66) /ó ìì/
 ó- ìì
 2PL- brain
 ‘your(pl) brain’

(67) /ó a íí/
 w -á íí
 2PL -POSS water
 ‘your(pl) water’

Note that the 2SG pronoun /i/ before inalienable possessa does not provide a comparable test: the homorganicity restriction on prevocalic glide formation (i.e., that V_1V_2 must differ in major place features) independently blocks the process before front-vowel-initial possessa regardless of prosodic structure (e.g., see [í-ìì] ‘your(sg) brain’ in Table 5).²³ The contrast between (66) and (67) thus supports the ω domain restriction on prevocalic glide formation. Because the pronoun vowel is a bare nucleus in both configurations, the difference in phonological outcomes is not attributable to syllabic structure. Rather, it tracks the prosodic word boundary.

The possessor data also bear on the intervocalic voiced obstruent restriction introduced in § 3.2. In the STAMP domain and with post-verbal particles, voiced obstruent onsets are absent in intervocalic position: the NPST auxiliary /be/ and the particle /-da/ surface without their onset after vowels. The possessor constructions provide a contrasting environment that sharpens the descriptive generalization.

Consider possessa with initial /b/. In the inalienable paradigm (Table 5), /bò/ ‘hand’ retains its onset across the full range of pronominal possessors, crucially including those with final vowels: 2SG [í-bò], 3SG [à-bò], 1DU [mò-bò], etc. In the alienable paradigm (Table 6), /bú/ ‘farm’ likewise retains its onset after the possessive morpheme /-a/: 1SG [n-á bú], 2SG [j-á bú], 3SG [à-à bú], and so forth. In both configurations, /b/ survives in an intervocalic position.²⁴ The relevant difference lies in prosodic structure. In the possessor constructions, both alienable and inalienable, the possessum heads its own prosodic word, and its onset is at the left edge of ω . The non-past auxiliary /be/ and the particle /-da/, however, do not head their own prosodic words and do not occur at the left edge of one. The descriptive generalization is that the intervocalic voiced obstruent restriction does not target onsets at the left edge of a prosodic word.

²³When pronouns ending in a nasal (1SG, 1EXCL, 3PL) occur before /ìì/ ‘brain’, the onset of the possessum surfaces with apparent palatalization, i.e., [-dʒìì], as shown in Table 5. This alternation is not analyzed here.

²⁴The asymmetry extends to the object-verb boundary. In (i), the verb *béjà* ‘fall’ retains its initial /b/ after the vowel-final case marker on the preceding object, which is predicted if the verb and object occur in distinct prosodic domains, separated by a phonological phrase boundary. Place assimilation of nasals, on the other hand, is not sensitive to this boundary: in (ii), the plural suffix /-N/ on the object assimilates to the onset of the following verb, surfacing as [m] before /bò/ ‘harvest’. This mirrors the broader pattern observed throughout § 3, where place assimilation operates across prosodic boundaries that the intervocalic voiced obstruent restriction respects.

(i) mùsú-n-á jàgbási-à *béjà*
 woman-PL-PST onion-OBJ fall
 ‘The women dropped the onion.’

(ii) mùsú-n-á jàgbási-m *bò*
 woman-PL-PST onion-PL harvest
 ‘The women harvested the onions.’

The possessor data thus extend the evidence for the proposed analysis in two directions. First, the same underlying forms and phonological processes established at the subject-auxiliary boundary produce the expected surface realizations in an independent syntactic domain. Second, the possessor configurations provide independent evidence for the prosodic domain distinctions proposed at the beginning of § 3: prevocalic glide formation is restricted to within a prosodic word, while the intervocalic voiced obstruent restriction does not target onsets at the left edge of a prosodic word. Together with the auxiliary-by-auxiliary evidence presented in §§ 3.1 to 3.3, these results indicate that the surface variation in the STAMP paradigm is systematic. This generality is predicted if the surface forms arise from the interaction of independently motivated phonological processes with concatenated exponents; it is unexplained if the STAMP forms are single, listed portmanteau exponents bearing no compositional relationship to the pronoun forms that appear elsewhere in the grammar.

4 Internal arguments and non-structural case

The parallelism argument in § 3 demonstrated that the phonological processes conditioning the pronominal STAMP paradigm also operate at the subject-auxiliary boundary with full DP subjects and at the possessor-possessum boundary. This section further extends the empirical domain. First, the same phonological processes are shown to operate on pronominal objects and case markers, further demonstrating that the patterns established in § 3 are not confined to the subject-auxiliary boundary (§ 4.1). Second, case marking provides a new intervention diagnostic: when case-marked internal arguments surface as derived subjects in unaccusatives, the case marker linearly intervenes between the subject and the auxiliary and systematically disrupts the phonological processes that would otherwise apply (§ 4.2).²⁵ Finally, § 4.3 shows that pronominal objects can condition the auxiliary’s phonological realization, with consequences that extend back to the subject-auxiliary boundary.

4.1 Non-structural case in Kono

Certain verbs in Kono assign an overt case marker to their internal argument, surfacing as a suffix on the argument whether it is a full DP or a pronoun. Two case forms are attested: *-a* and *-ɔ*. The choice between them is lexically determined by the verb. Some verbs assign *-a*, (68); others assign *-ɔ*, (69); and many verbs assign no overt marker at all, (70).²⁶

- (68) n-á kájánè-n-à tó
 1SG-PST door-PL-**OBJ** close
 ‘I closed the doors.’

²⁵The focus marker and the case marker appear to be the only sources of intervention at the subject-auxiliary boundary available in Kono. This contrasts with languages like Gã, where phrasal constituents such as adverbial PPs can intervene between the subject pronoun and the auxiliary (Felice 2021, 2022; Russell 2024).

²⁶The three-way split is lexically specific and not predictable from verb class, thematic roles, or definiteness/specificity—a hallmark of lexical case cross-linguistically (Woolford 2006). Historically, these markers are reflexes of the West Mande postpositions **la* (locative) and **ro* (inessive) (Kastenholz 1996, cf. Vydin 2016), functional elements argued to originate from relational nouns that grammaticalized into adpositional and case-marking heads (Claudi 1994; Heine and Kuteva 2004). Synchronically, these morphemes exhibit the hallmark distribution of Kono postpositions (occurring post-nominally) but function as case suffixes on internal arguments.

- (69) n-á mǐdʒì-n-dɔ dùù
 1SG-PST needle-PL-OBJ bend
 'I bent the needles.'
- (70) m-bé jàgbási-n tèè-à
 1SG-NPST onion-PL cut-PRT
 'I am slicing these onions.'

The case marker /-a/ surfaces invariably as [a]. In (71a), it follows the plural suffix /-N/, which surfaces as default coronal [n] before vowels, and in (71b), it attaches directly to the vowel-final singular noun.

- (71) a. /sàni N a/
 bɔ̀ndú-á sàni -n -à tó
 Bondu-PST bottle -PL -OBJ close
 'Bondu closed the bottles.'
- b. /sàni a/
 bɔ̀ndú-á sàni -à tó
 Bondu-PST bottle -OBJ close
 'Bondu closed the bottle.'

The case marker /-dɔ/ exhibits a phonological alternation conditioned by the preceding segment. After the plural nasal in (72a), the onset /d/ surfaces and the nasal assimilates to its place. After the vowel-final singular noun in (72b), the onset /d/ is absent and only the vowel [ɔ] surfaces. This alternation is attributed to the same intervocalic voiced obstruent alternation identified for NPST /be/ and the post-verbal particle /-da/ in § 3.2, now extended to yet another morpheme in a distinct morphosyntactic environment.

- (72) a. /sàni N dɔ/
 bɔ̀ndú-á sàni -n -dɔ tè
 Bondu-PST bottle -PL -OBJ break
 'Bondu broke the bottles.'
- b. /sàni dɔ/
 bɔ̀ndú-á sàni -ɔ tè
 Bondu-PST bottle -OBJ break
 'Bondu broke the bottle.'

The case markers also attach to pronominal objects, where the phonological properties of pronouns give rise to additional interactions predicted by the analysis. The onset alternation of /-dɔ/ established for DP objects holds equally for pronominal objects. In (73), the 3PL pronoun's final nasal precedes the case marker, the onset /d/ surfaces, and the nasal assimilates, while in (74), the 2SG pronoun /í/ conditions the absence of the onset /d/.

- (73) /àN dɔ/
 n-á àn -dɔ tè
 1SG-PST 3PL -OBJ break
 'I broke them.'

- (74)
- | | | | |
|----------------|-----|------|-------|
| | /í | dɔ/ | |
| n-á | j | -ɔ́ | tè |
| 1SG-PST | 2SG | -OBJ | break |
| 'I broke you.' | | | |

In (74), with the case marker's onset absent, 2SG /í/ is now adjacent to the case marker vowel /ɔ/. Since these vowels differ in major place features, the conditions for prevocalic glide formation are met, and the surface form is [j-ɔ́]. As with the NPST auxiliary in § 3.2, the intervocalic voiced obstruent restriction feeds prevocalic glide formation.

Glide formation is also observed with the /-a/ case marker, where the conditions are met directly: in (75), 2SG /í/ → [j] before /a/—the same realization observed before PST (cf. (28)) and the alienable possessive morpheme /-a/ (cf. § 3.4).

- (75)
- | | | | |
|-----------------|-----|------|-------|
| | /í | a/ | |
| n-á | j | -á | tó |
| 1SG-PST | 2SG | -OBJ | close |
| 'I closed you.' | | | |

The application of prevocalic glide formation in (74) and (75) reflects the same prosodic property of pronouns established throughout § 3: pronouns do not form independent prosodic words; when the pronoun combines with a case marker, the two are parsed into a single prosodic word. The resulting prosodic structure, sketched in (76), parallels those proposed for the subject-auxiliary boundary and possessor domains.

- (76) (ϕ (ω pro case))

When the object is a full DP, the DP forms its own prosodic word and the case marker falls outside it, as sketched in (77) (cf. (21)).

- (77) (ϕ (ω DP) case)

The prosodic structure in (77) is proposed by hypothesis; the onset alternation of /-dɔ/ is consistent with it, but is not independently diagnostic, since the intervocalic voiced obstruent restriction applies both within and across ω boundaries (§§ 3.2 and 3.4).

4.2 Case retention in unaccusatives

The case markers established in the preceding section appear not only on objects of transitive verbs, but also on the sole argument of certain unaccusative constructions. In Kono, certain verbs assign case to their internal arguments in transitive constructions, and in inchoative unaccusatives built from these verbs, the case marker is retained on the internal argument when it surfaces in subject position. The case marker thus surfaces on the derived subject, which crucially precedes the auxiliary in the linear string—a configuration that places the case marker between the subject and the auxiliary.

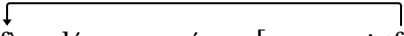
This pattern is illustrated with the verb *tè* 'break', which assigns the /-dɔ/ case marker. In the transitive in (78), the case-marked object 'these bottles' appears in its canonical post-auxiliary, pre-verbal position. In the unaccusative counterpart in (79), the same case-marked DP appears in the pre-auxiliary subject

position, with the case marker intact. The same pattern holds with pronominal internal arguments: in (80), the 3PL pronoun bears the case marker in subject position before the auxiliary.

- (78) bòndú -ní sàni-tfɛ-n-dó tè-à,...
 Bondu -PROG bottle-DET-PL-OBJ break-PRT
 'Bondu was breaking these bottles,...'
- (79) sàni-tfɛ-n-dó -ní tè-à,...
 bottle-DET-PL-OBJ -PROG break-PRT
 'These bottles were breaking,...'
- (80) àn-dó -ní tè-à,...
 3PL-OBJ -PROG break-PRT
 'They were breaking,...'

The retention of the case marker under A-movement conforms to a well-established diagnostic for non-structural case: structural case is typically lost under movement to a derived position, while non-structural case (whether inherent, lexical, quirky, etc.) is retained (Woolford 2006; Zaenen and Maling 1990, among others). The Kono pattern is consistent with this diagnostic. The case marker is lexically assigned by the verb, surfaces identically on objects in transitives and on derived subjects in unaccusatives, and is not lost under A-movement. While the primary evidence for the subject status of the moved argument comes from its pre-auxiliary linear position (the same position occupied by subjects across all other clause types examined in this paper), it is at least consistent with an analysis in which A-movement targets Spec,IP (for EPP or related reasons).²⁷

The unaccusative structure with a derived subject is schematized in (81), where the case-marked DP originates as the complement of V and moves to the pre-auxiliary subject position.

- (81)  sàni-tfɛ-n-dó -ní [_{VP} <sani-tfɛ-n-dó> tè-à]...
 bottle-DET-PL-OBJ -PROG break-PRT
 'These bottles were breaking...'

The syntactic configuration in (81) has a direct phonological consequence. In § 3, the phonological processes conditioning the surface forms of the STAMP paradigm were shown to be sensitive to linear adjacency at the subject-auxiliary boundary. When the case marker intervenes in this position, it disrupts that adjacency. The final segment of the subject proper is no longer adjacent to the auxiliary; instead, the case marker is adjacent to the auxiliary. The phonological processes that applied at the

²⁷ Additional diagnostics for unaccusativity include incompatibility with agent-oriented instrumental PPs, as in (ib), and compatibility with agent-denying 'by itself', as in (iib). Since these examples use the past, the unaccusatives do not show overt reflexes of the past auxiliary (see § 4.2.3).

- | | |
|---|--|
| <p>(i) a. n-á mèmè-ò tè kóné à
 1SG-PST mirror -OBJ break stick P
 'I broke the mirror with a stick.'</p> <p>b. mèmè-ò tè-à (*kóné à)
 mirror-OBJ break-PRT (*stick P)
 'The mirror broke (*with a stick).'</p> | <p>(ii) a. kái_k-à mèmè-ò_j tè à-wándì_{k/ɔj} kùrà
 Kai-PST mirror-OBJ break 3SG-REFL by
 'Kai_k broke the mirror_j by him/itself_{k/ɔj}.'</p> <p>b. mèmè-ò_j tè-à à-wándì_{j/ɔk} kùrà
 mirror-OBJ break-PRT 3SG-REFL by
 'The mirror_j broke by itself_{j/ɔk}.'</p> |
|---|--|

subject-auxiliary boundary in § 3 are therefore predicted to be blocked, with new processes, conditioned by the case marker, applying in their place.

This prediction is borne out already with DP subjects. Consider the verb *túné* ‘sink’, which assigns the /-a/ case marker. The transitive construction in (82) provides a baseline with a case-marked direct object, *gbèngbéè-n-à* ‘boats’.

- (82) *m* -bé *gbèngbéè-n-à* *túné-á*
 1SG -NPST boat-PL-OBJ sink-PRT
 ‘I am sinking the boats.’

In the unaccusative in (83), the DP surfaces in subject position, and its case marker now intervenes between the plural nasal and the NPST auxiliary.

- (83) /*gbèngbéè-N-a* *be*/
 gbèngbéè-n-è -è [_{VP} *t* *túné-á*]
 boat-PL-OBJ -NPST sink-PRT
 ‘The boats are sinking.’

The phonological outcome in (83) differs strikingly from what would be expected without the case marker. The plural nasal surfaces as [n] (its default coronal realization before a vowel), rather than assimilating to [m] as it would before the labial /b/ of the auxiliary. And the auxiliary surfaces without its onset /b/, since it is now preceded by the vowel of the case marker rather than the nasal. The auxiliary and the case marker exhibit the bidirectional vowel assimilation observed elsewhere when an /a/ vowel immediately precedes NPST /be/ (cf. § 3.2). In short, the presence of the case marker bleeds assimilation of the plural suffix to [m] while also feeding the intervocalic voiced obstruent restriction for the auxiliary.

That this contrast is indeed due to the intervening case marker, and not to some other structural difference between transitives and unaccusative constructions, is supported by comparing (83) with the unergative in (84). Here, a plural DP subject precedes the non-past auxiliary with no intervening case marker. The plural nasal assimilates to [m] and the auxiliary surfaces as [-be]—precisely the pattern predicted when adjacent.

- (84) /*mùsú-N* *be*/
 mùsú-m -bé *túné-tjà-à*
 woman-PL -NPST sink-LV-PRT
 ‘The women are diving.’

The contrast between (83) and (84) isolates the case marker as the critical variable. Both involve a plural DP subject preceding the NPST auxiliary; the crucial difference here is the presence of the intervening case marker in the unaccusative. Where the case marker is present, the plural/nasal-NPST/bilabial adjacency is disrupted and phonological processes are blocked. Where the case marker is absent, the processes apply as in § 3. As we will see in the following sections, the general pattern observed with DP subjects extends straightforwardly to pronominal subjects, as predicted by the decomposed analysis.

4.2.1 Progressive auxiliary

The progressive auxiliary provides the most straightforward illustration of intervention effects, because its onset is invariant across all environments. There is no onset alternation to complicate the picture;

the only adjacency-sensitive processes at the subject-auxiliary boundary occur with a preceding nasal, as established in § 3.3. When a case marker intervenes, these processes are correctly predicted to be blocked.

Recall that 1SG /ŋ/ coalesces with the nasal onset of the progressive auxiliary when the two are adjacent, yielding a single surface nasal [ní] that bears the high tone of the pronoun, as shown in (85), repeated from § 3.3. No segmental trace of the boundary between the pronoun and the auxiliary remains on the surface.

- (85) /ŋ ní/
 Ø -ní káŋáà-à tón-dà, mbɛ bondu na
 1SG -PROG door-OBJ close-PRT, then Bondu come
 ‘I was closing the door, then Bondu came.’

When the 1SG pronoun bears a case marker in an unaccusative construction, coalescence is blocked. With the /-a/ case marker, the surface form is [n-á-ní], as in (86a); with /-dɔ/, the surface form is [n-dɔ-ní], as in (86b).

- (86) [CONTEXT: *Fictional story where inanimate objects (doors, bottles) report what happened to them.*]
 a. /ŋ a ní/
 n -á -ní tón-dà, mbɛ bondu na
 1SG -OBJ -PROG close-PRT, then Bondu come
 ‘I was closing, then Bondu came.’
 b. /ŋ dɔ ní/
 n -dɔ -ní tè-à, mbɛ bondu na
 1SG -OBJ -PROG break-PRT, then Bondu come
 ‘I was breaking, then Bondu came.’

The case marker disrupts the adjacency between the pronoun nasal and the auxiliary onset, and the pronoun, case marker, and auxiliary surface as three distinct exponents.

The 3PL pronoun /àn/ shows a parallel pattern. Its final nasal coalesces with the progressive auxiliary onset when the two are adjacent, shown in (87).

- (87) /àn ní/
 à -ní káŋáà-à tón-dà,...
 3PL -PROG door-OBJ close-PRT
 ‘They were closing the door,...’

When a case marker intervenes, coalescence is again blocked. With the /-a/ case marker in (88a), the surface form is [àn-à-ní]; with /-dɔ/ in (88b), the surface form is [àn-dɔ-ní].

- (88) a. /àn a ní/
 àn -à -ní tón-dà,...
 3PL -OBJ -PROG close-PRT
 ‘They were closing,...’

- b. /àN dɔ ni/
 àn -dɔ -ní tè-à,...
 3PL -OBJ -PROG break-PRT
 ‘They were breaking,...’

The pronoun’s final nasal, which leaves no trace on the surface in (87), now surfaces overtly as [n], interacting phonologically with the case markers rather than the auxiliary.

The 3SG pronoun /à/ illustrates a distinct facet of the intervention pattern. When 3SG bears the /-dɔ/ case marker, the case marker’s onset /d/ is absent after the pronoun vowel, as expected from the intervocalic voiced obstruent restriction. The resulting vowel sequence coalesces to a single segment: /à-dɔ/ → à-ɔ → [ɔ̃]. The back round quality is diagnostic of the case marker’s underlying vowel, since the pronoun alone is /à/. This coalescence is shown in the transitive in (89), where [ɔ̃] appears as the object. In the unaccusative in (90), the same [ɔ̃] appears in subject position before the auxiliary, which surfaces identically in both configurations.

- (89) /à dɔ/
 à-nì ɔ̃ tè-à
 3SG-PROG 3SG.OBJ break-PRT
 ‘He was breaking it.’

- (90) /à dɔ ni/
 ɔ̃ -ní tè-à,...
 3SG.OBJ -PROG break-PRT
 ‘It was breaking,...’

The progressive data thus establish the case marker intervention pattern in its cleanest form. The auxiliary itself is segmentally stable, and the only apparent variable is whether the case marker intervenes between the subject and the auxiliary.

4.2.2 Non-past auxiliary

The non-past auxiliary presents a richer set of intervention effects, as both the auxiliary’s onset alternation and its vowel quality are affected by the change in linear adjacency. The full DP evidence for this was already presented in the comparison between (83) and (84) above. With pronominal subjects, the same intervention effects appear, and the consequences for the surface form of the STAMP paradigm are particularly striking.

When the 1SG pronoun /ń/ is adjacent to NPST /be/, the two yield [ń-bé], as in (91). That this surface form is recoverable as the concatenation of two distinct underlying exponents is a consequence of the analysis developed in § 3.2.

- (91) /ń be/
 ń -bé káńánà-à tón-dà wá
 1SG -NPST door-OBJ close-PRT FOC
 ‘I will close the door.’

When a case marker intervenes in unaccusatives, the surface forms change dramatically. With the /-a/ case marker in (92), the surface form is [n-é-é]; with the /-dɔ/ case marker in (93), it is [n-dɔ-ɔ̃].

(92) /Ń a be/
 n -é -é tón-dà wǎ́
 1SG -OBJ -NPST close-PRT FOC
 ‘I will close.’

(93) /Ń dɔ be/
 n -dɔ -ɔ tè-à wǎ́
 1SG -OBJ -NPST break-PRT FOC
 ‘I will break.’

The contrast between [m-bé] in (91) and [n-é-é]/[n-dɔ-ɔ] in (92) and (93) reflects the case marker restructuring the linear adjacency relations at the subject-auxiliary boundary. Without a case marker, the pronoun nasal is adjacent to the auxiliary onset; place assimilation and onset realization proceed as in § 3.2. With a case marker, the case marker’s vowel is adjacent to the auxiliary onset. The case marker bleeds place assimilation to [m] (the nasal is no longer adjacent to the labial onset) and feeds the intervocalic voiced obstruent restriction (the auxiliary onset is now preceded by a vowel rather than a nasal). The resulting hiatus between the case marker and the auxiliary is resolved through vowel assimilation: in (92), /-a-be/ → -a-e → [ɛ-ɛ]; in (93), /-dɔ-be/ → -dɔ-e → [dɔ-ɔ]. In both cases, the auxiliary vowel adjusts to the preceding vowel, though the /ae/ sequence shows the bidirectional accommodation observed elsewhere (cf. (43)), whereas the /ɔe/ sequence shows full assimilation of the auxiliary vowel to the case marker.

The 2SG pronoun provides a complementary illustration. In (94), 2SG /í/ + NPST /be/ yields [é-é], with glide formation blocked by the homorganicity restriction (§ 3.2) and hiatus resolved through assimilation.

(94) /í be/
 é -é káṇánà-à tón-dà wǎ́
 2SG -NPST door-OBJ close-PRT FOC
 ‘You will close the door.’

When a case marker intervenes, the vowel adjacent to the pronoun is no longer the front vowel /e/ but the case marker vowel (/a/ or /ɔ/), and glide formation proceeds. With the /-a/ case marker in (95), the surface form is [j-á-é]; with the /-dɔ/ case marker in (96), we have [j-ɔ-ɔ].

(95) /í a be/
 j -á -é tón-dà wǎ́
 2SG -OBJ -NPST close-PRT FOC
 ‘You will close.’

(96) /í dɔ be/
 j -ɔ -ɔ tè-à wǎ́
 2SG -OBJ -NPST break-PRT FOC
 ‘You will break.’

The 2SG data thus complement the 1SG pattern: where the case marker bleeds the processes that would otherwise apply with 1SG (place assimilation, onset realization), it also feeds glide formation with 2SG by removing the homorganic environment that otherwise blocks it.

The 3SG pronoun /à/ with the /-dɔ/ case marker illustrates the same intervention logic. Without the case marker, 3SG /à/ + NPST /be/ yields [è-è] (cf. (43)). With the /-dɔ/ case marker intervening, as in (97), the pronoun and case marker coalesce to [ɔ̃] (cf. (90)), and it is this [ɔ̃] that is now adjacent to the auxiliary. The auxiliary onset is again absent, and the auxiliary vowel assimilates to the quality of the preceding segment, yielding [ɔ̃-ɔ̃]. The contrast between [è-è] and [ɔ̃-ɔ̃] thus reflects a change in what is adjacent to the auxiliary vowel.

- (97) /à dɔ be/
 ɔ̃ -ɔ̃ tè-à wá
 3SG.OBJ -NPST break-PRT FOC
 ‘It will break.’

The intervention effects illustrated with the 1SG, 2SG, and 3SG pronouns extend to the remaining person/number categories in predictable ways.

4.2.3 Past auxiliary

The picture with the past auxiliary is less transparent than with the progressive and non-past. In transitive constructions, PST /a/ surfaces right-adjacent to the subject, as in (98).

- (98) bɔndú -á sàni -ò tè
 Bondu -PST bottle -OBJ break
 ‘Bondu broke the bottle.’

In unaccusatives, however, the case marker is clearly present on the sole argument, but the auxiliary does not appear to surface overtly.²⁸ This is shown with the /-dɔ/ case marker in (99) and (100). In (99), the DP *mèmè* ‘mirror’ surfaces with [-ò], diagnostic of the case marker. In (100), the 3PL pronoun surfaces with the case marker as [àn-dò], as expected.

- (99) mèmè -ò tè-à
 mirror -OBJ break-PRT
 ‘The mirror broke.’
- (100) àn -dò tè-à
 3PL -OBJ break-PRT
 ‘They broke.’

In neither case is there any reflex of PST /a/—no additional vowel, no lengthening, no tonal trace beyond what is expected from the argument and case marker alone. This contrasts with NPST in (101), where the auxiliary loses its onset and its vowel assimilates to the preceding vowel.

- (101) àn -dò -ɔ̃ tè-à wá
 3PL -OBJ -NPST break-PRT FOC
 ‘They will break.’

²⁸ The post-verbal particle in the unaccusative examples (e.g., *tè-à* in (99)) is not a reflex of past tense. This particle, whose underlying form is /-da/, appears in both past and non-past constructions (see (44) and (45)). Its distribution is orthogonal to tense marking.

With verbs assigning the /-a/ case marker, the picture is predictably more ambiguous. The surface forms in (102) and (103) could contain both the case marker and a coalesced auxiliary vowel, since both are underlyingly /a/.²⁹ Whether the auxiliary is simply absent rather than coalesced with the case markers remains an open question.

(102) kájánè-n -á tónd-à
 door-PL -OBJ close-PRT
 ‘The doors closed.’

(103) àn -à tón-dà
 3PL -OBJ close-PRT
 ‘They closed.’

A further complication is that without an overt reflex of the auxiliary, the pre-auxiliary position diagnostic for A-movement is unavailable. While the case marker’s presence is consistent with a derived subject analysis (as established for PROG and NPST above), the PST data alone cannot confirm it. The intervention argument accordingly rests on the PROG and NPST evidence, where the auxiliary surfaces overtly and the phonological effects of case marker intervention are unambiguous. The PST data are indeed consistent with this analysis, but, at face value, they do not directly strengthen it.

One further question left open by the evidence presented in § 4.2 concerns the prosodic parse of the case-marked pronoun and auxiliary. In § 4.1, pronominal objects with a case marker were argued to be parsed within a prosodic single word: (ω pro case). Whether the auxiliary is also contained within this prosodic word in the unaccusative configuration, e.g., (ω pro case aux), or falls outside it is not resolved by the data examined here. The vowel assimilation between the case marker and the auxiliary vowel observed in § 4.2.2 is consistent with a single- ω parse, but does not require it, since comparable assimilation was shown to occur across ω boundaries elsewhere in the language (cf. (42) in § 3.2). The intervention argument developed in this section does not depend on this distinction. What is critical is that the change in segmental adjacency and the phonological alternations it gives rise to are evidence for morphosyntactic decomposability, regardless of the prosodic relationship between the case marker and the auxiliary.

²⁹ More broadly, every instance of vowel coalescence with mora loss in the data examined in this paper involves /a/ as at least one of the participating vowels, whether the 3SG pronoun /à/, the /-a/ case marker, or (potentially) PST /a/. When NPST is involved, the mora is consistently preserved and only assimilation occurs (cf. /-dɔ-be/ → -dɔ-e → [-dɔ-ɔ] in (101), where the case marker and auxiliary vowel each retain a distinct timing slot).

This asymmetry is consistent with coalescence of PST /a/ with an adjacent vowel, but it does not rule out the alternative that PST is simply absent in these constructions. The latter possibility may connect to another pattern in Kono: several intransitive verbs show no overt past tense marking at all, with the subject immediately preceding the verb, as in (i) and (ii). Whether these constructions reflect a phonologically null PST allomorph, a tonal realization of the auxiliary, or a distinct syntactic configuration is an open question that requires further investigation.

(i) m bɛjà kúnú
 1SG fall yesterday
 ‘I fell yesterday.’

(ii) n sémbè-à
 1SG stop-PRT
 ‘I stopped.’

4.3 Phonological conditioning from pronominal objects

The phonological processes established at the subject-auxiliary boundary can also operate between the auxiliary and a following pronominal object. Under certain conditions, the auxiliary's surface form is conditioned not only by the subject that precedes it but also by the object that follows it, when that object is a pronoun. More strikingly, this rightward conditioning can feed processes at the subject-auxiliary boundary itself: the vowel quality of a pronominal object can condition the auxiliary vowel, which in turn alters the environment for the preceding subject pronoun. Under a portmanteau analysis, such sensitivity to the object is unexpected, since the subject-auxiliary form would be determined prior to insertion. Under the decomposed analysis adopted here, it follows straightforwardly: the auxiliary is an independent exponent, and its phonological realization responds to adjacent segments in both directions. If the subject pronoun, auxiliary, and pronominal object are parsed into a single prosodic word, as the data examined below suggest, the resulting domain is broader than any proposed in § 3, grouping elements from distinct syntactic projections, none of which form a syntactic constituent. The implications of this for the mapping between syntactic structure and prosodic domains are addressed briefly in § 5.

Consider the verb *tè* 'break', which assigns the /-dɔ/ case marker. When the direct object is 3SG /à/, the pronoun and the case marker coalesce to [ɔ̃] (cf. (89)). This [ɔ̃] conditions the vowel quality of the *preceding* auxiliary. In (104), the past auxiliary /a/ surfaces as [ɔ̃], assimilating to the back round quality of the following [ɔ̃]. The same assimilation holds with the non-past auxiliary in (105): the onset [b] is intact after the 1SG nasal, but the vowel assimilates to [ɔ̃] rather than surfacing as [é].

(104) /Ñ a à dɔ/
 n -ɔ̃ ɔ̃ tè
 1SG -PST 3SG .OBJ break
 'I broke it.'

(105) /Ñ be à dɔ/
 m -bɔ̃ ɔ̃ tè-à wá
 1SG -NPST 3SG .OBJ break-PRT FOC
 'I will break it.'

The vowel interaction is blocked, however, when a consonantal segment intervenes. In (106), for instance, glide formation applies to the case-marked 2SG direct object: /í-dɔ/ → í-ɔ̃ → [j-ɔ̃]. The resulting glide [j] intervenes between the auxiliary vowel and the object vowel, and the auxiliary surfaces as [bé] with its vowel unaffected by the pronominal object (cf. (105)).

(106) /Ñ be í dɔ/
 m -bé j -ɔ̃ tè-à wá
 1SG -NPST 2SG -OBJ break-PRT FOC
 'I will break you.'

Auxiliary vowel assimilation thus applies when the auxiliary vowel and the pronominal object vowel are linearly adjacent, and is blocked when a consonantal segment intervenes. Notably, while the glide formation that produces the intervening [j] in (106) itself occurs at the boundary between the object pronoun and the case marker, the domain of application is consistent with the single prosodic word parse offered above.

The most striking consequence of the auxiliary vowel assimilation involves its effect on the preceding subject pronoun. When the 2SG pronoun /í/ is the subject and a pronominal object conditions the auxiliary vowel, the change in the auxiliary's vowel quality feeds prevocalic glide formation at the subject-auxiliary boundary. In (107), the auxiliary vowel assimilates to the following 3SG object, surfacing as [ɔ́] rather than [é]. This, in turn, removes the front-vowel environment that blocks glide formation via the homorganicity restriction (§ 3.2), and glide formation applies: /í/ → [j]. The contrast with (108) isolates the pronominal object as the critical variable: with a full DP object, the auxiliary does not assimilate, and the 2SG subject surfaces as [é], not as [j].

(107) /í be a do/
 j -ɔ́ ɔ́ tè-à wá́ sínà
 2SG -NPST 3SG .OBJ break-PRT FOC tomorrow
 'You(sg) will break it tomorrow.'

(108) é -é mèmè -ɔ́ tè-à wá́ sínà
 2SG -NPST mirror -OBJ break-PRT FOC tomorrow
 'You(sg) will break the mirror tomorrow.'

The glide formation observed in (107) has a further implication for prosodic structure. Throughout § 3 and § 4.2, glide formation was argued to apply within a single prosodic word but not across a prosodic word boundary. If this domain restriction is correct, the data suggest that the subject pronoun, auxiliary, object pronoun, and case marker are parsed into a single prosodic word: (_ω pro₁ aux pro₂ case).

If this parse is correct, the feeding relationship in (107) should not depend on the specific vowel quality [ɔ́] of the object. The same pattern is predicted whenever a pronominal object's vowel conditions the auxiliary *and* thereby removes the homorganic environment. This prediction is borne out with the verb *jà* 'prank', which does not assign a case marker. In (109), the bare 3SG object /à/ conditions the auxiliary vowel to surface as [á] rather than [é], and glide formation of the 2SG subject applies. In (110), with a full DP object, no assimilation occurs and 2SG-NPST surfaces as [é-é].

(109) /í be à/
 j -á à já-à wá́ sínà
 2SG -NPST 3SG prank-PRT FOC tomorrow
 'You will prank her tomorrow.'

(110) /í be/
 é -é bòndú já-à wá́ sínà
 2SG -NPST Bondu prank-PRT FOC tomorrow
 'You will prank Bondu tomorrow.'

The evidence in § 4 extends the decomposed analysis beyond the subject-auxiliary boundary. Pronominal objects and case markers give rise to the same phonological interactions observed elsewhere, and when a case marker linearly intervenes between the subject and the auxiliary, the processes established in § 3 are systematically disrupted. The conditioning from pronominal objects further demonstrates that the auxiliary's phonological realization can be determined by adjacent segments in both directions, with consequences that propagate to the subject-auxiliary boundary itself. These patterns follow straightforwardly from a model in which pronouns, case markers, and auxiliaries are distinct exponents whose surface realization is determined by general phonological processes sensitive to the prosodic structures in which these exponents are parsed.

5 Concluding remarks

This paper has presented the first detailed morphosyntactic analysis of the Kono STAMP paradigm. The central claim defended here is that the surface variation in Table 1 is phonologically derived rather than suppletive: the underlying forms proposed for the pronouns and the auxiliaries derive the attested surface forms through independently motivated phonological processes. After establishing that subject-referencing morphemes in the paradigm are pronouns occupying an argument position (§ 2), I showed that the same inventory of phonological processes conditions the surface realization of both full DPs and pronouns, with differences in realization tracking the distinct prosodic structures associated with each (§ 3). § 4 extended the empirical domain to pronominal objects and case markers, showing that the same phonological processes operate in these configurations as predicted by the analysis. The case-marking data provided a further diagnostic: when a case marker linearly intervenes between the subject and the auxiliary, these phonological processes are systematically disrupted in ways predicted by changes in linear adjacency. The evidence from pronominal objects (§ 4.3) extends this logic beyond the STAMP domain, showing that the auxiliary's phonological realization is conditioned by adjacent segments in both directions, with consequences that can propagate back to the form of the subject pronoun itself. That this sensitivity to adjacency and prosodic structure holds across these disparate configurations is expected if pronouns, auxiliaries, and case markers are distinct exponents interacting phonologically; it is unexplained if STAMP forms are single portmanteau exponents.

The intervention evidence also bears on the structural analysis of non-structural case in Kono. The case markers /-a/ and /-dɔ/, situated in § 4.1 as synchronic reflexes of historically reconstructed West Mande postpositions (cf. Kastenholz 1996; Vydrin 2016), occupy a distinct position in the linearized string between the subject and the auxiliary in unaccusatives. This is consistent with an analysis in which the case marker heads its own functional projection, though the phonological argument developed here does not depend on this specific structural implementation. What the argument requires is that the case marker constitutes a distinct exponent that participates in the phonological processes that apply to the concatenated string, and the data directly support this minimal requirement.

The theoretical consequences of these findings are direct. As established in § 1, the question of which morphological mechanism derives a portmanteau paradigm (e.g., Fusion, Spanning, etc.) arises only when suppletion has been established. Since the Kono paradigm has been shown not to be suppletive, these mechanisms are inapplicable: there are no portmanteaux to derive. The pronoun and the auxiliary remain separate terminal nodes targeted by separate Vocabulary Items, and the phonological component bears the full burden of deriving the surface paradigm from concatenated underlying forms. Each terminal in the paradigm investigated here appears to have a single underlying exponent, with no contextual allomorphy at the level of individual morphemes. It bears emphasizing that this conclusion does not constitute an argument against Fusion as an operation available in morphological theory; the claim is rather that the empirical precondition for invoking such an operation (i.e., a genuinely suppletive paradigm) is simply not met in Kono.

The analysis developed here also has implications for the mapping between syntactic structure and prosodic domains. Throughout § 3 and § 4, subject pronouns and auxiliaries are parsed into a single prosodic word despite not forming a syntactic constituent: a subject pronoun occupies a specifier position and the auxiliary heads some extended verbal projection. The data in § 4.3 suggest an even broader prosodic domain. If the hypothesis that a subject pronoun, auxiliary, and pronominal object are parsed into a single ω is correct, then a single prosodic word includes elements on both sides of the auxiliary, no subset of which forms a syntactic constituent. Prosodic domain formation in Kono is thus

not straightforwardly derivable from syntactic constituency, and the nature of the mapping between syntactic and prosodic structure remains an open question that the empirical investigation here is well positioned to inform.

Several aspects of the Kono system warrant further investigation. The interactions between the tonal system and the segmental phonology documented here remain to be analyzed in full detail; the tonal specifications of individual morphemes and the tonal processes operating within and across the prosodic structures proposed here are likely to provide additional insight into the analysis. The present analysis has also focused exclusively on tense/aspect marking in affirmative constructions. The polarity dimension of the STAMP paradigm (i.e., the “P” in STAMP) has not been examined here and constitutes a natural next step for this line of research.

The Kono findings are situated within a growing body of work examining the derivational transparency of STAMP paradigms across the Macro-Sudan Belt. [Garvin et al. \(2021, To appear\)](#) observe that such paradigms vary considerably in the degree to which surface forms are phonologically predictable, and that many languages exhibit at least some transparent forms within an otherwise opaque-looking paradigm. [Russell \(2024\)](#) demonstrates that the vast majority of Gã STAMP forms are in fact predictable from regular phonological processes, though a few exceptional forms resist phonological derivation and may involve morpheme-specific suppletive allomorphy—precisely the forms that [Felice \(2021, 2022\)](#) derives via post-syntactic Fusion. The present study reaches a different conclusion for Kono: the phonological processes independently established for the language derive the full STAMP paradigm examined here, with no exceptional forms requiring *morphological* operations.

The methods developed here offer a generalizable toolkit for assessing derivational transparency in other STAMP paradigms. The point in § 1 about the burden of proof bears reiterating in this broader context: establishing suppletion requires demonstrating not merely that surface forms *appear* irregular or that they *can* be captured by post-syntactic restructuring operations, but that no motivated phonological derivation relates them. This in turn underscores the importance of investigating the morphosyntactic environments beyond the paradigm itself. The phonological processes that derive a STAMP paradigm may only become visible when one examines how the same morpheme behaves in other syntactic configurations (e.g., different subject types, different case-marking environments, etc.). The present study illustrates both the methods and the analytical stakes involved in meeting that burden.

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